

Systematic Design for a Successive Approximation ADC

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Outline

- Background
- Principles of Operation
- System and Circuit Design
- Case Study
 - Simulations
 - Layout Generation
 - Performance Evaluation
- Conclusion
- Perspectives

The Successive Approximation ADC

« The Return »

Moderate
Resolution

Low
Power

Minimum
Active blocs

Reconfigurable

- Emerging new Applications

- MEMS Sensor Interface:

- Resolution: 7-8 bits, BW=50kHz [Scott 2003]*

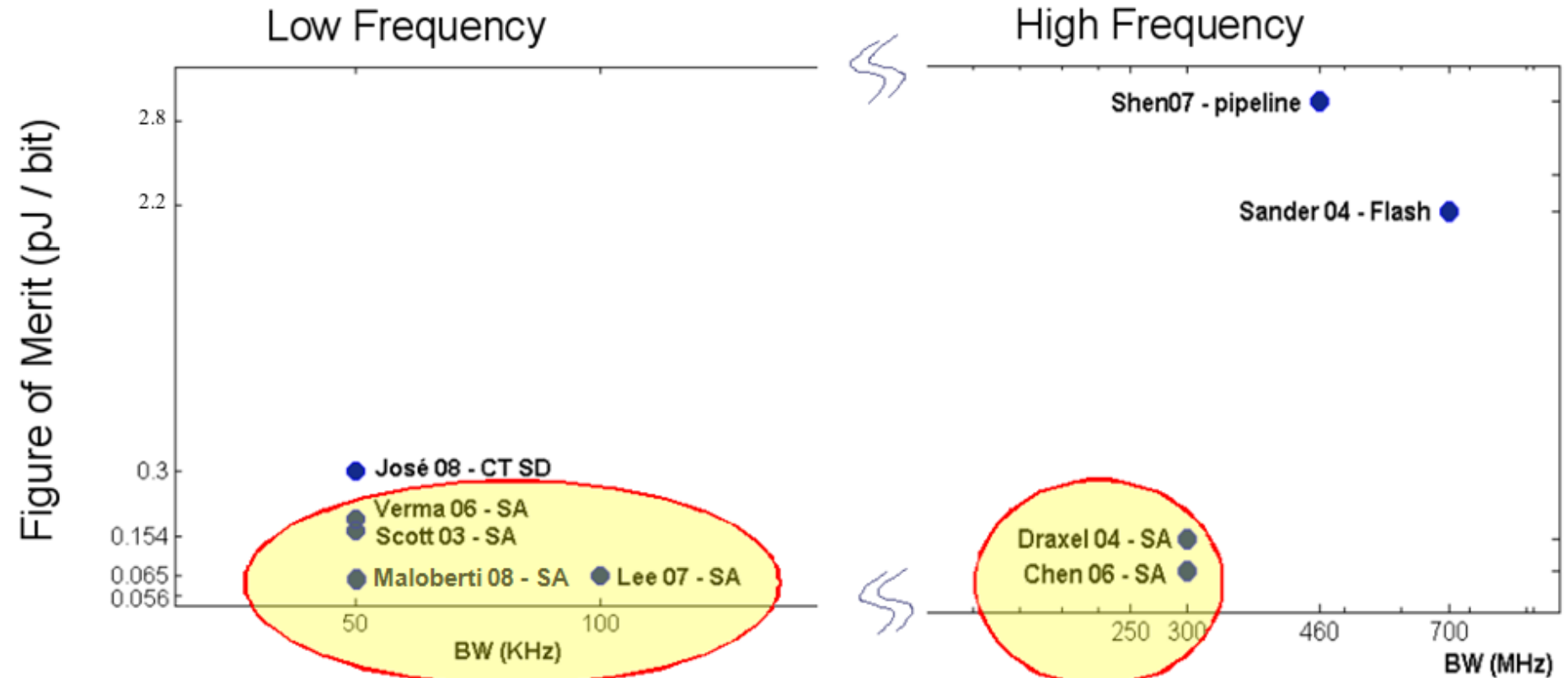
- Multi-standards RF receiver

- Resolution: 8 bits, BW = 20 MHz [Montaudon 2008]*

- Ultra Wide Band (wireless UWB):

- Resolution: 5-6 bits, BW=300MHz [Chen 2006]*

Figure of Merit



$$FOM = \frac{P}{2^{\text{Resolution}} * 2 * BW}$$

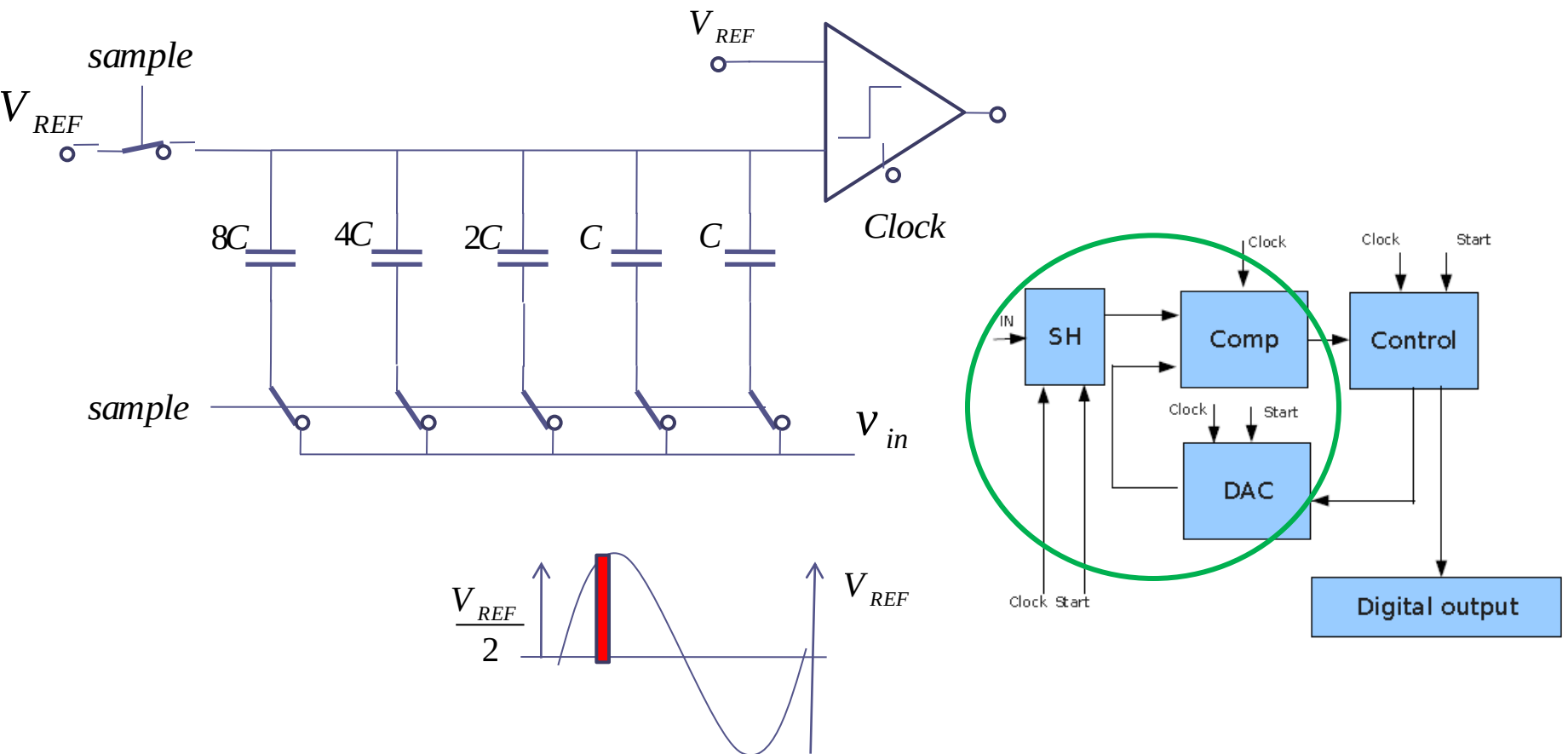
Objectives

- Develop a systematic design method for successive approximation ADC from system to layout level .
- Develop a general simulation environment with different levels of abstraction and programmed performance analysis.
- Emphasis on analog design automation and reuse techniques:
 - Automatic sizing
 - Layout generation
- Optimizing Layout for best matching

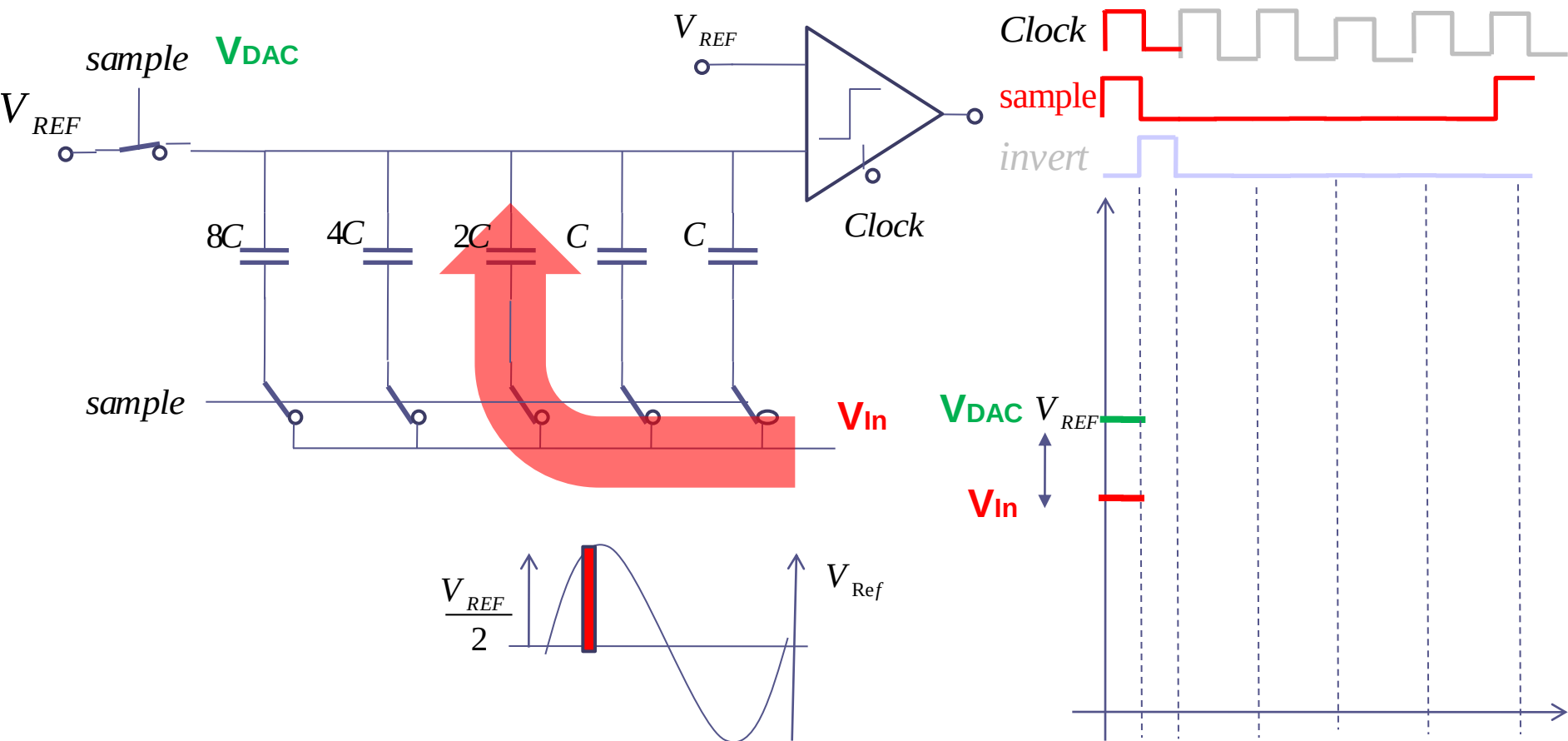
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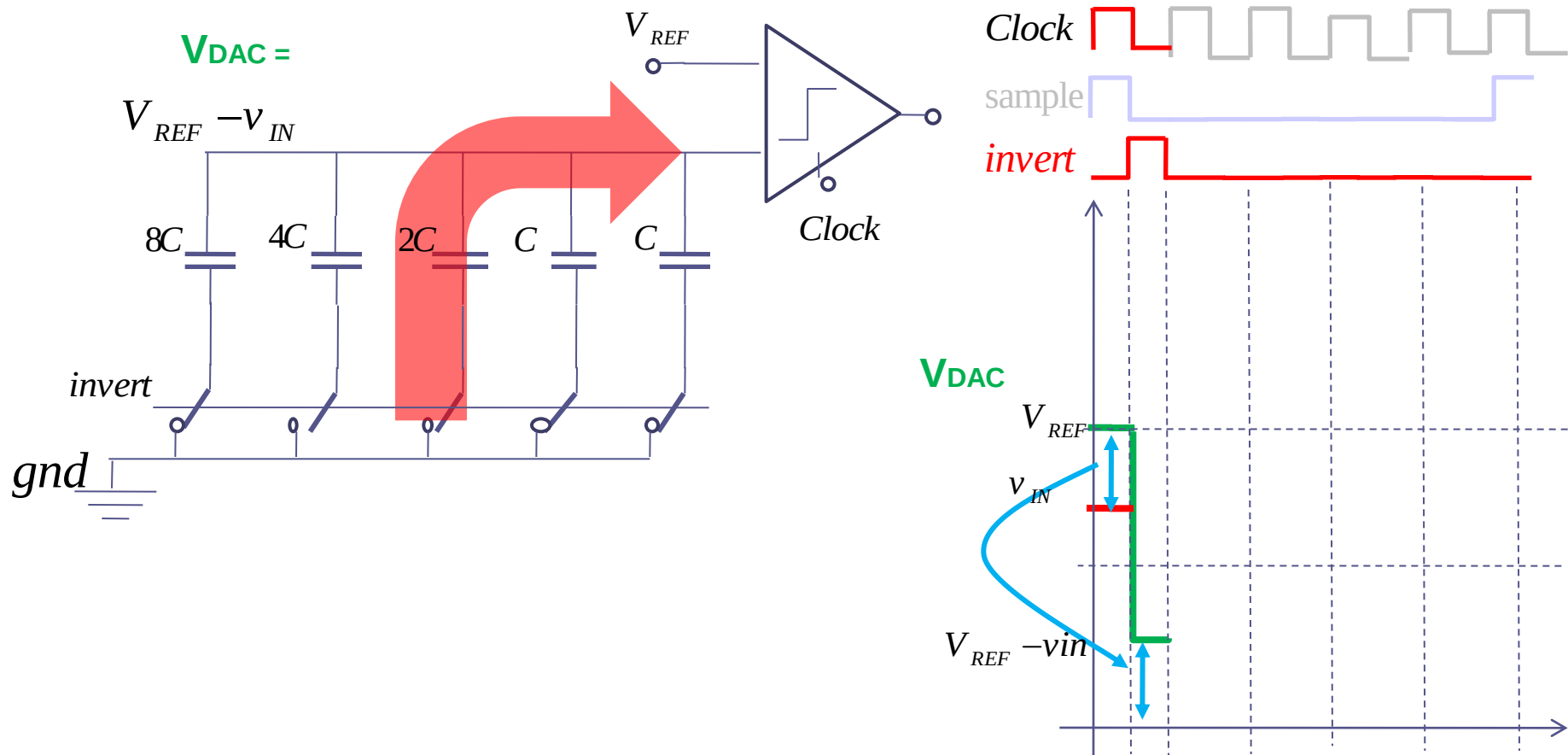
Single Ended SAR-ADC.



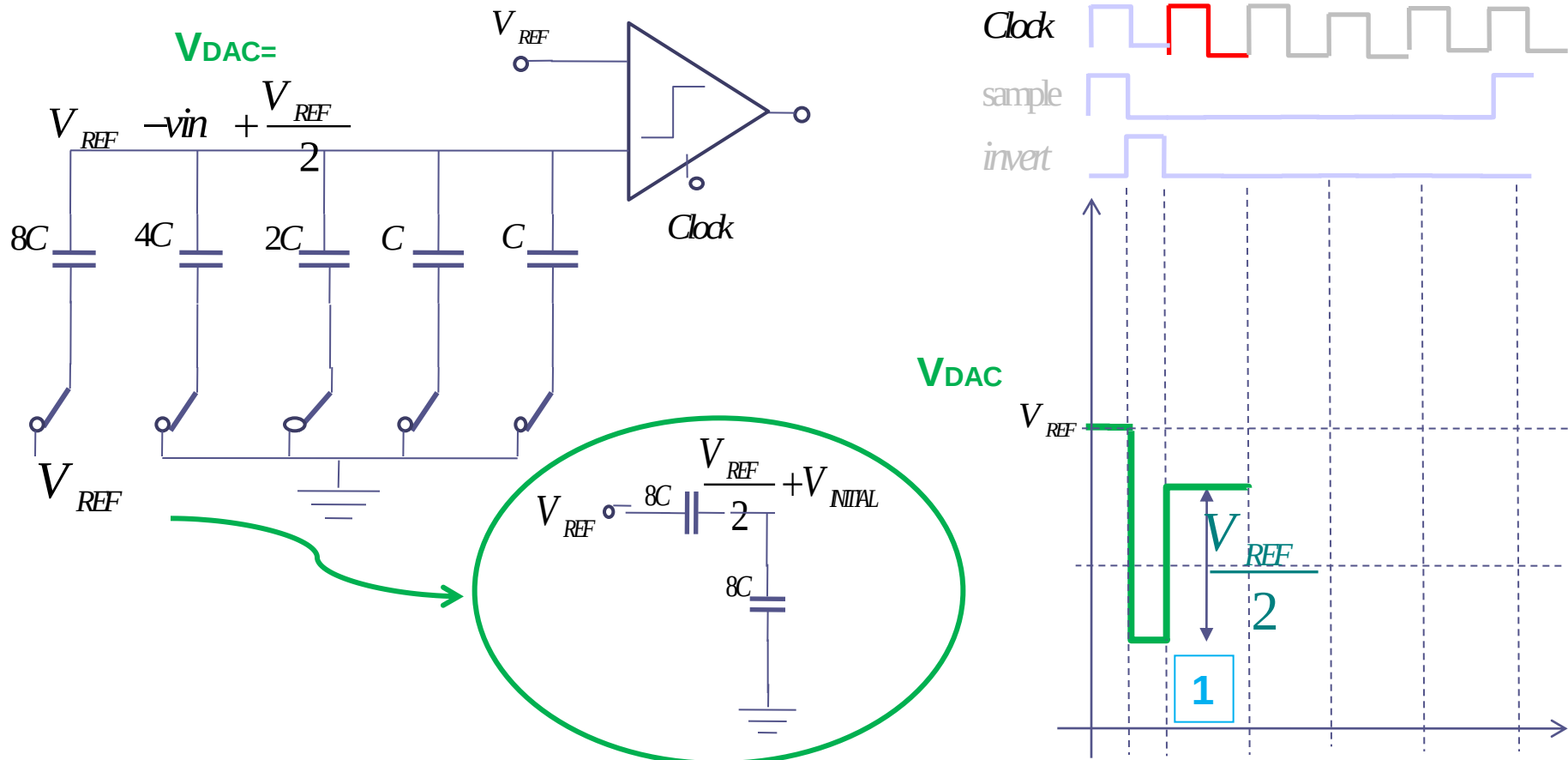
Sampling Mode



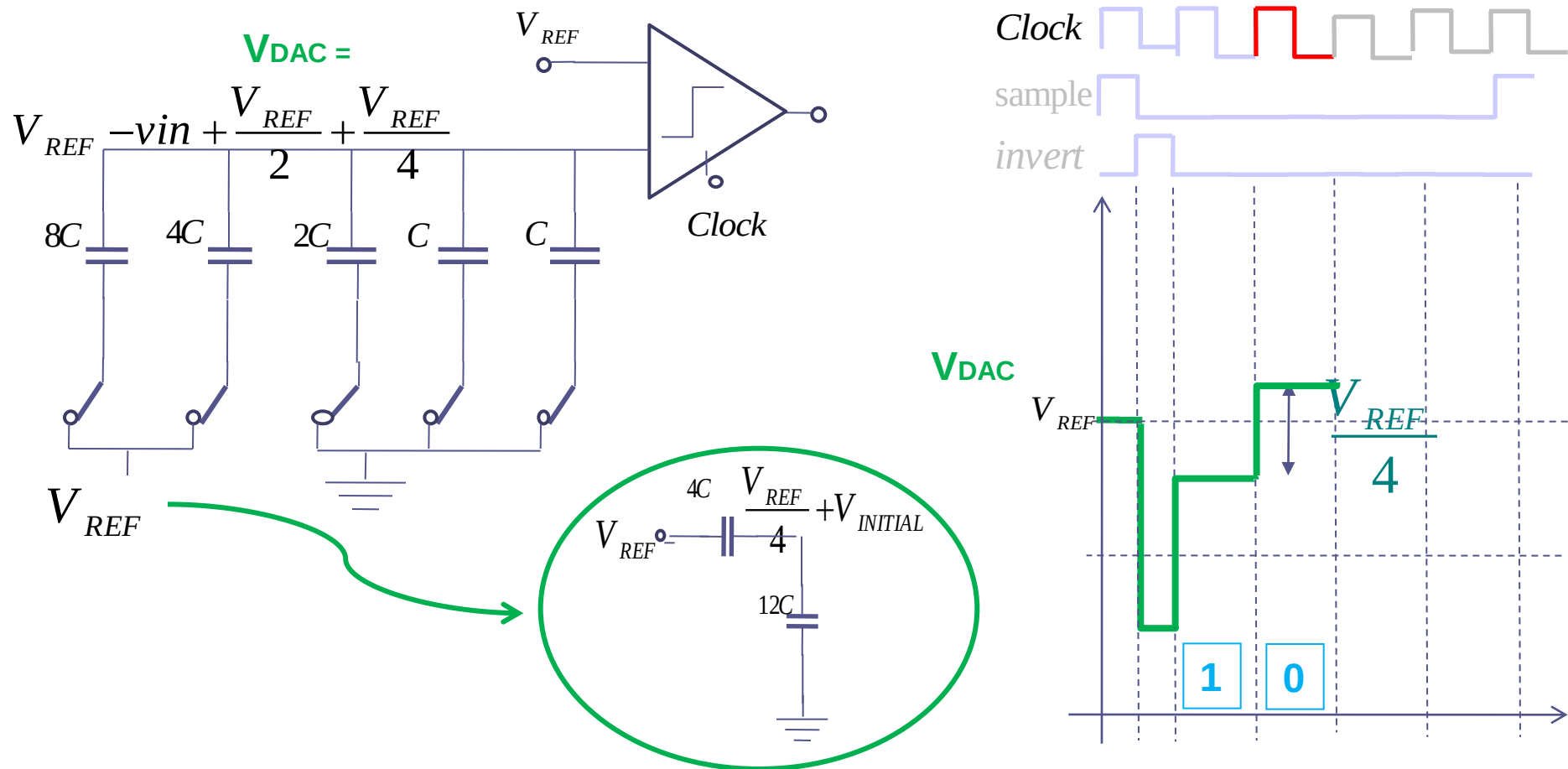
Inversion Mode



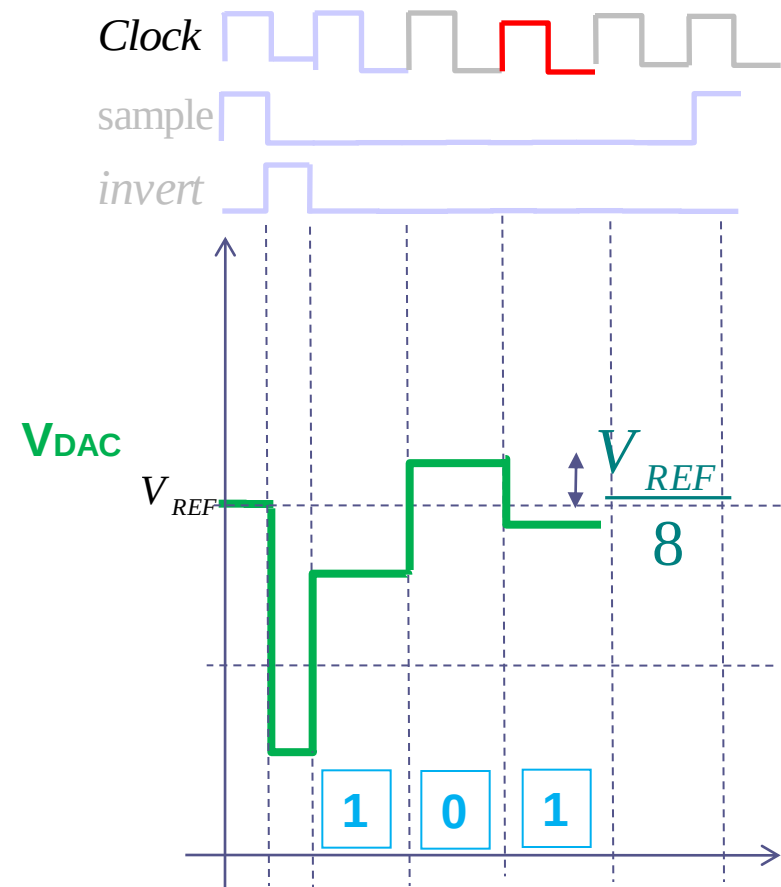
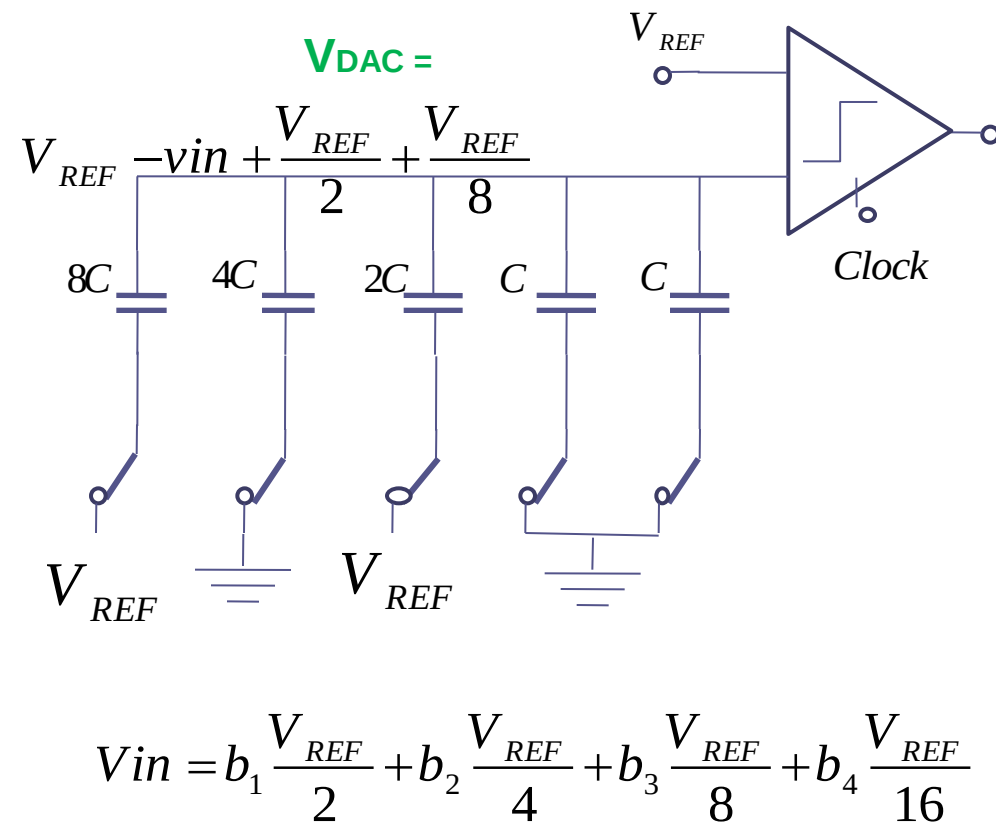
Charge redistribution mode (MSB)



Charge redistribution mode (MSB-1)

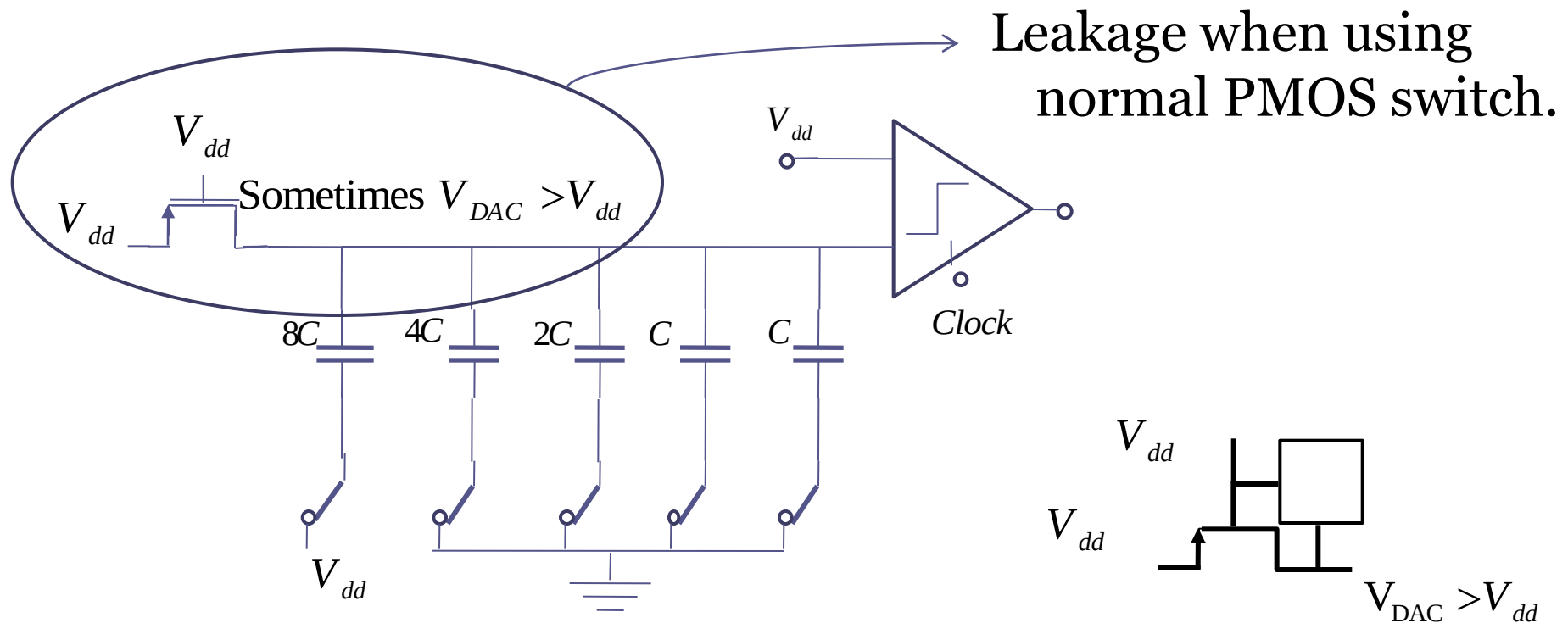


Mode Redistribution de la charge (MSB-2)



Problem

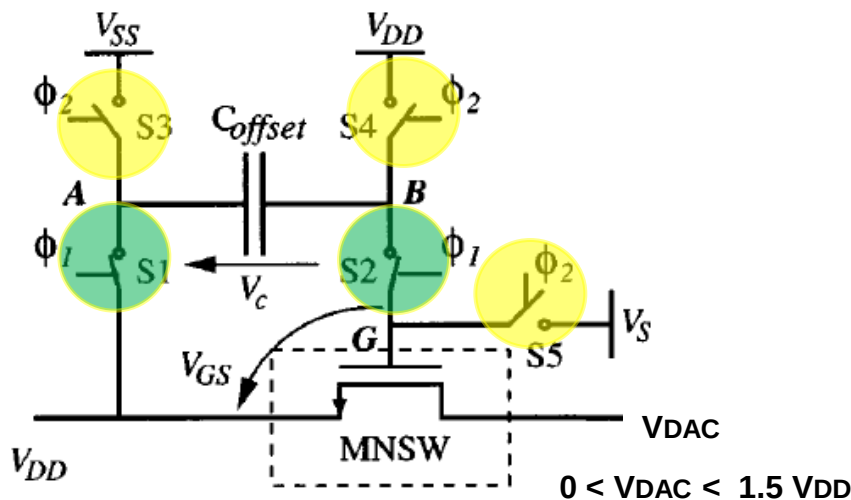
Selecting $V_{REF} = V_{dd}$ To increase the dynamic range



Possible Solution: Switched *charge-pump* [scotto3] or *Bootstrap* [dessouky01]

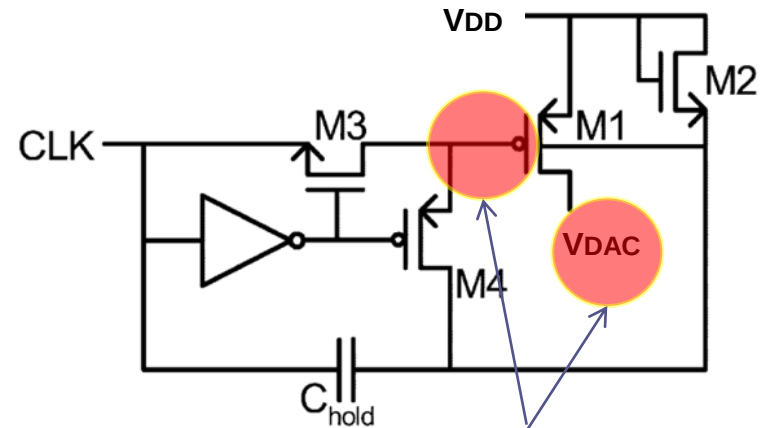
Possible solutions - Leakage

Bootstrap Switch

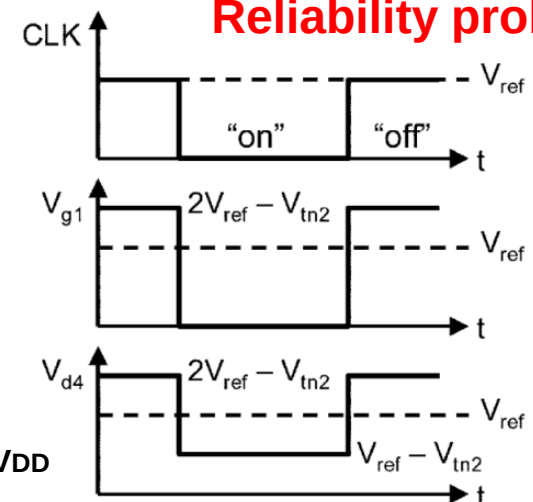


Since sometimes V_{DAC} value = $1.5 V_{DD}$
While $V_{G1} = 0$ when $M1$ is ON, $V_{GD,M1}$ exceeds V_{DD}

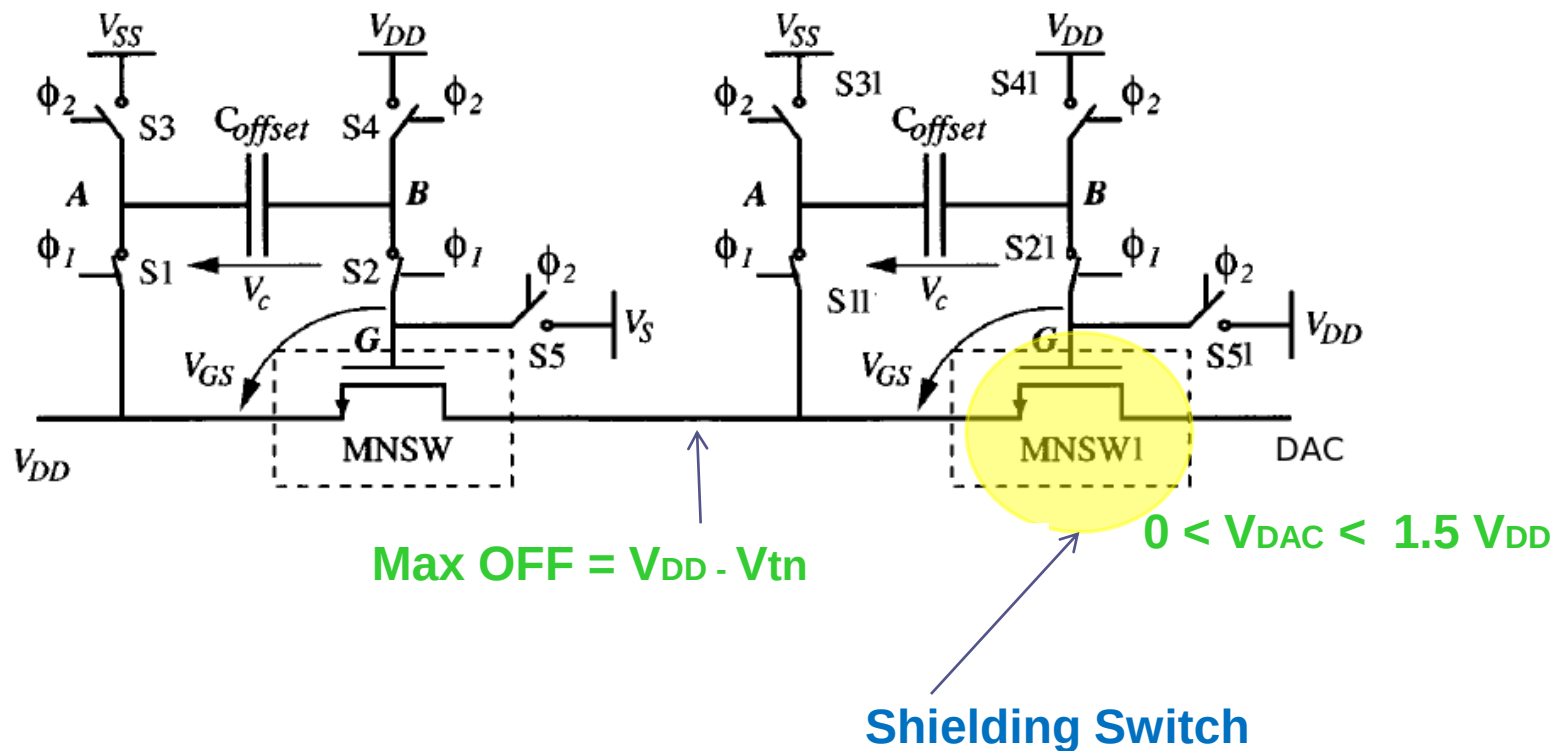
Charge pump Switch



Reliability problem

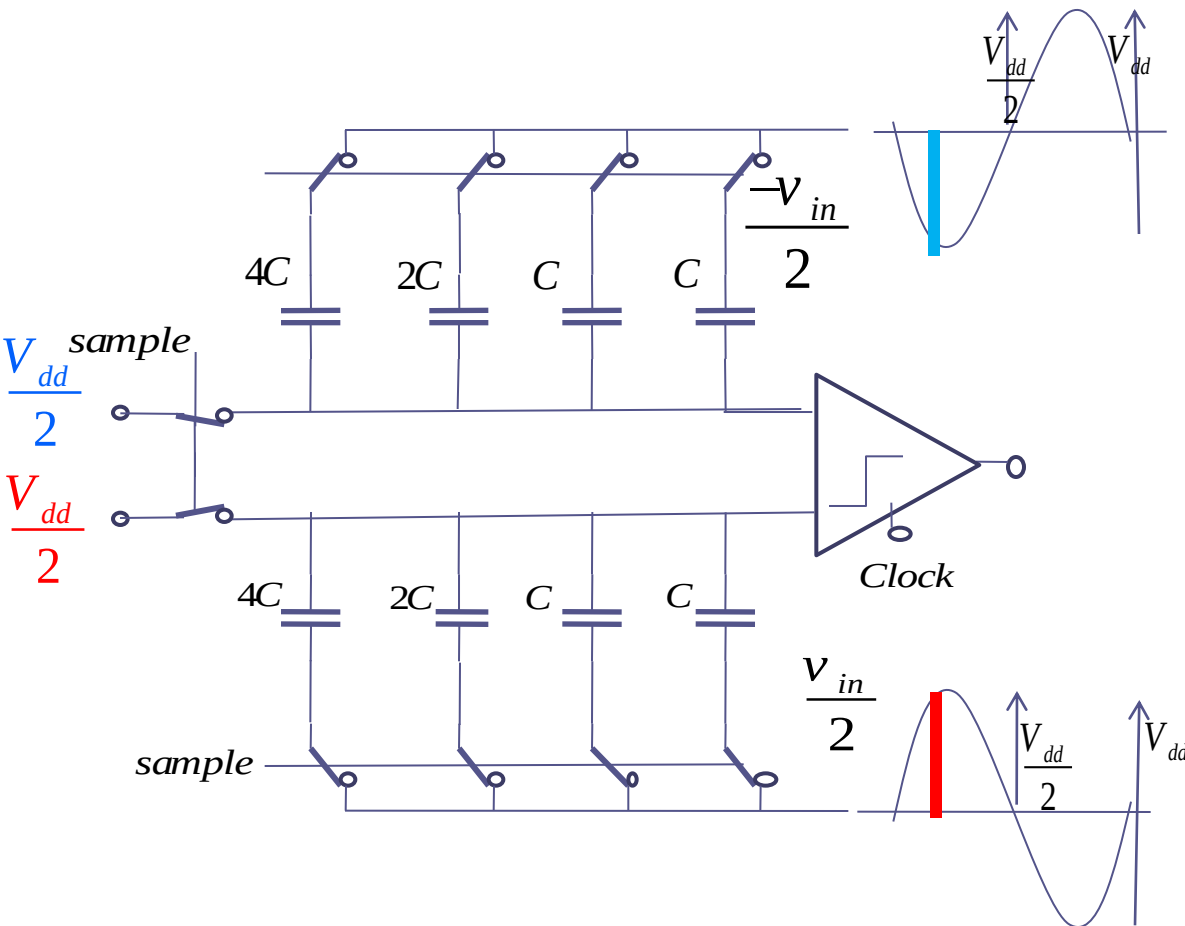


Possible solutions - Reliability



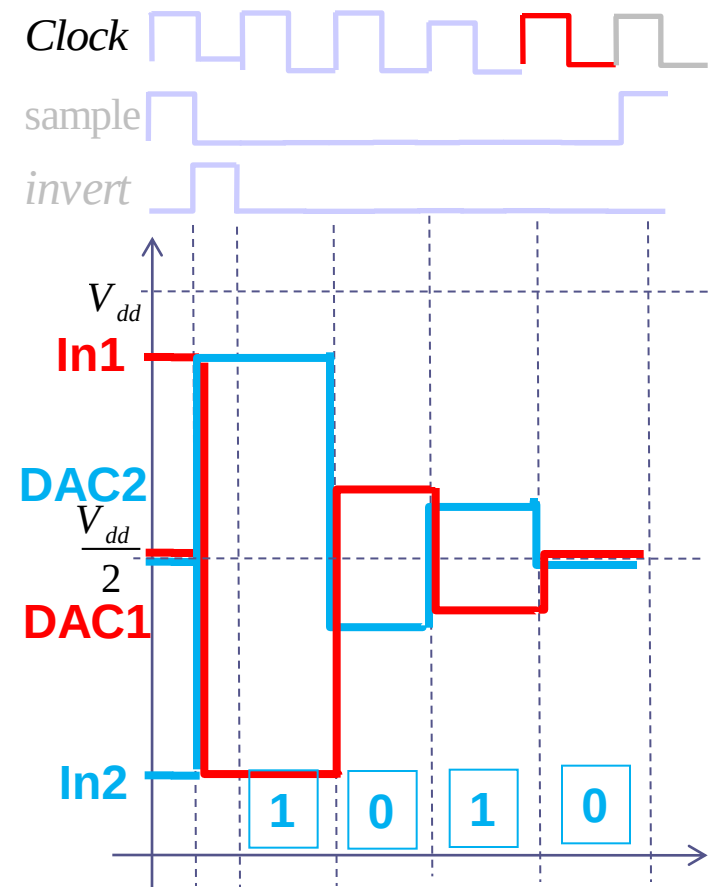
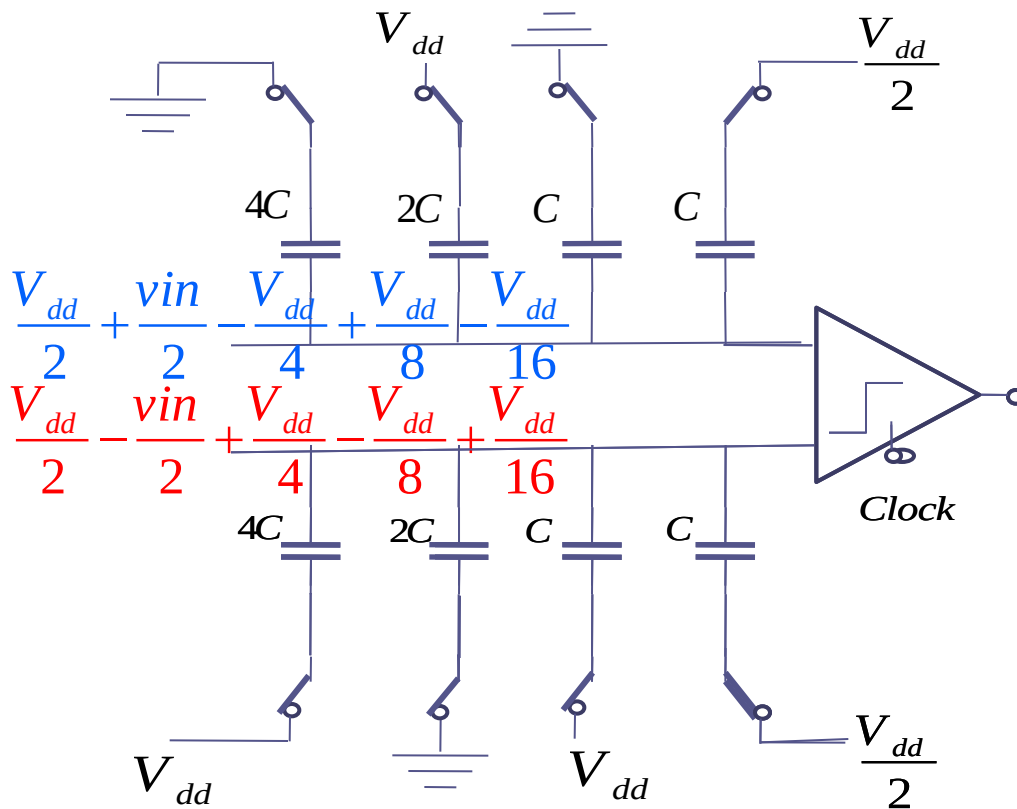


Differential SAR-ADC



Triple Reference

Differential SAR-ADC



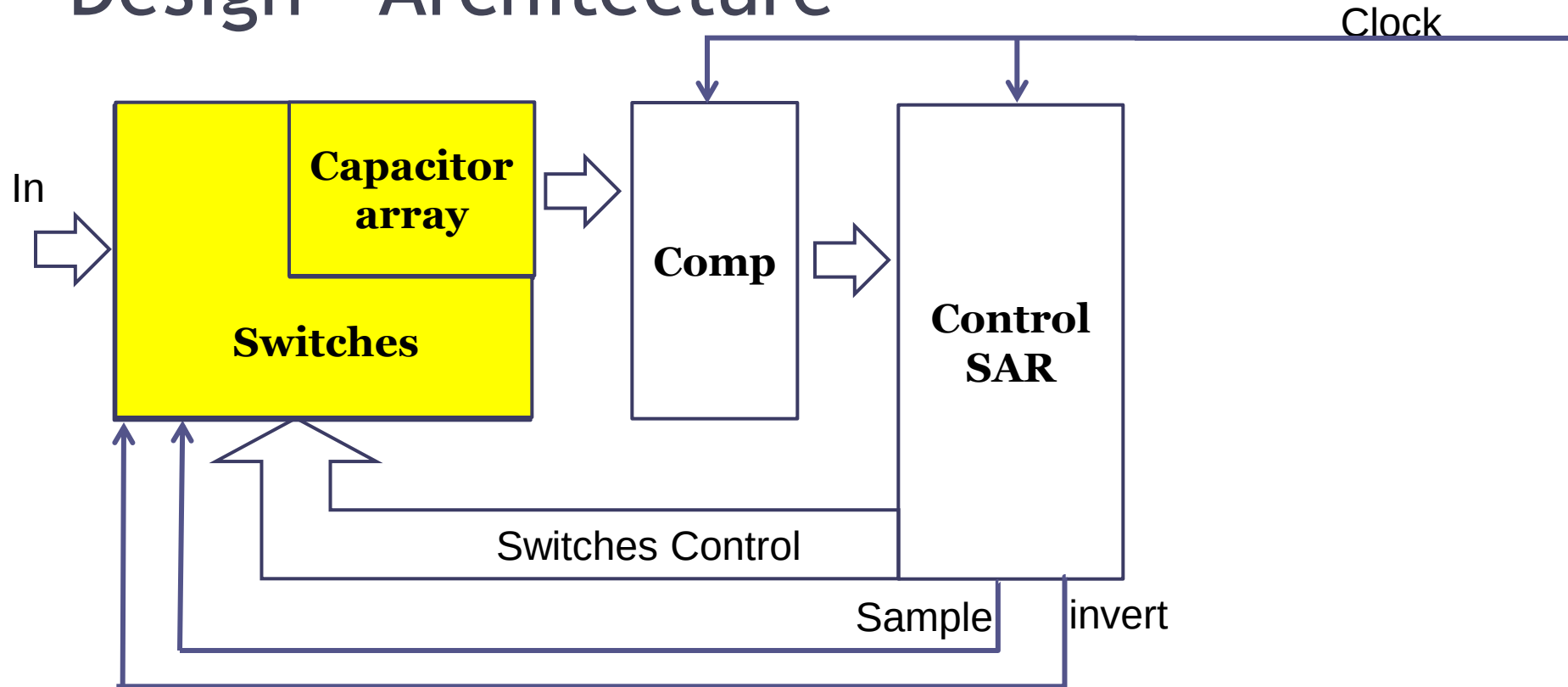
Operation - Summary

Single Ended Double reference	Differential Triple reference
	2 times the numbers of capacitors 6 times the numbers of switches
Special Switch (<i>charge-pump - bootstrap</i>)	No need for special switch
	Differential architectures advantages : - Suppressing even harmonics - Common mode rejection - Offset removal
Lower power consumption	Better performance at high frequencies

Outline

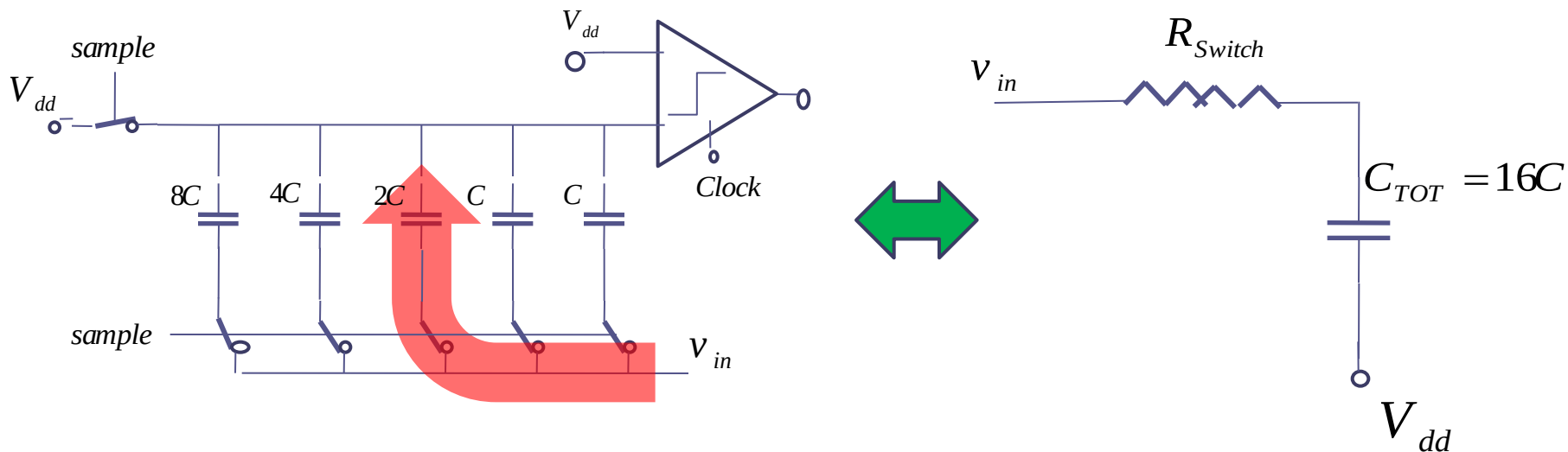
- Background
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- **System and Circuit Design**
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Design - Architecture



Capacitor array design issues - Noise

1- Thermal noise $\left[\frac{kT}{C_{TOT}} \right]$, due to Sampling

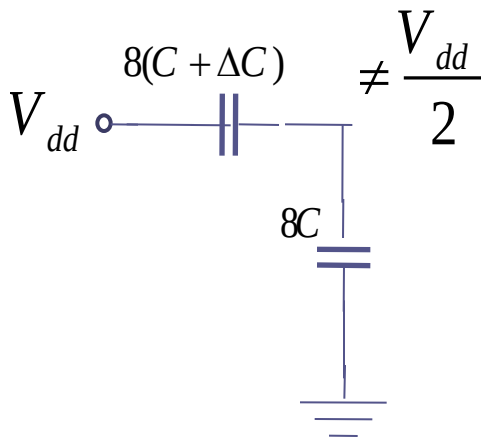


C_{Unit} increases, thermal noise decreases

Capacitor array design issues - Mismatch

2 – Capacitor Mismatch (Introduced in fabrication)

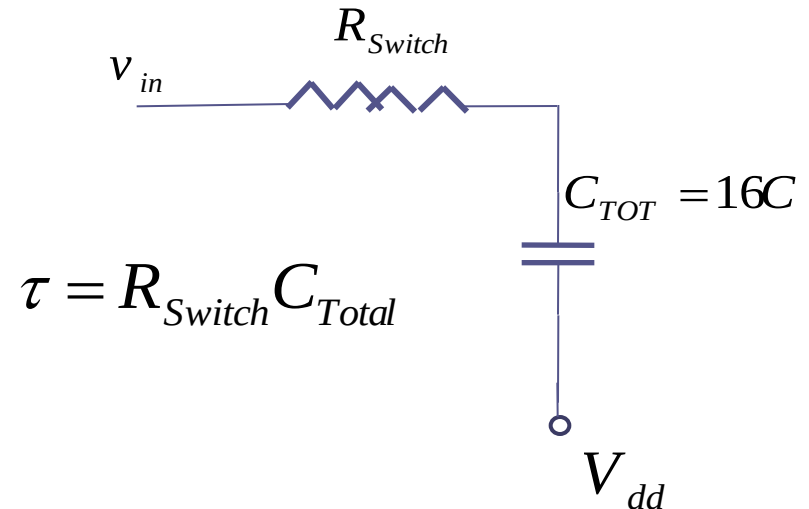
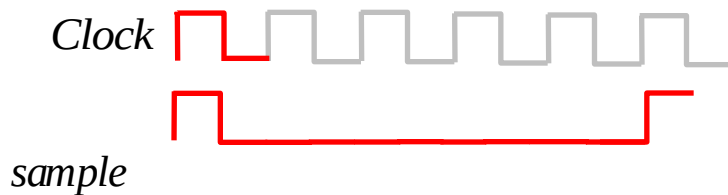
- Affects Generated comparison levels of the capacitive DAC



C_{Unit} increases, mismatch effect decreases

Capacitor array design issues - f_{Sampling}

3- Sampling Frequency



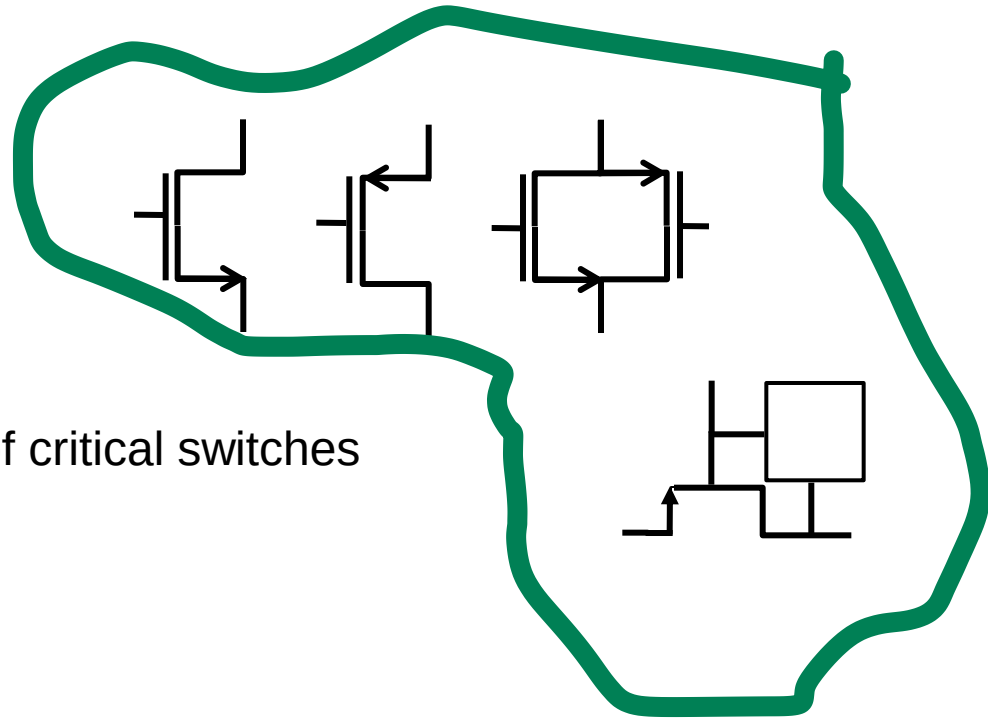
$$t_{\text{sampling}} \approx \frac{T_{\text{Clock}}}{2} \gg \tau \quad \text{For an accurate sampling}$$

C_{Unit} decreases, bandwidth increases

Switches

1) Switches selection

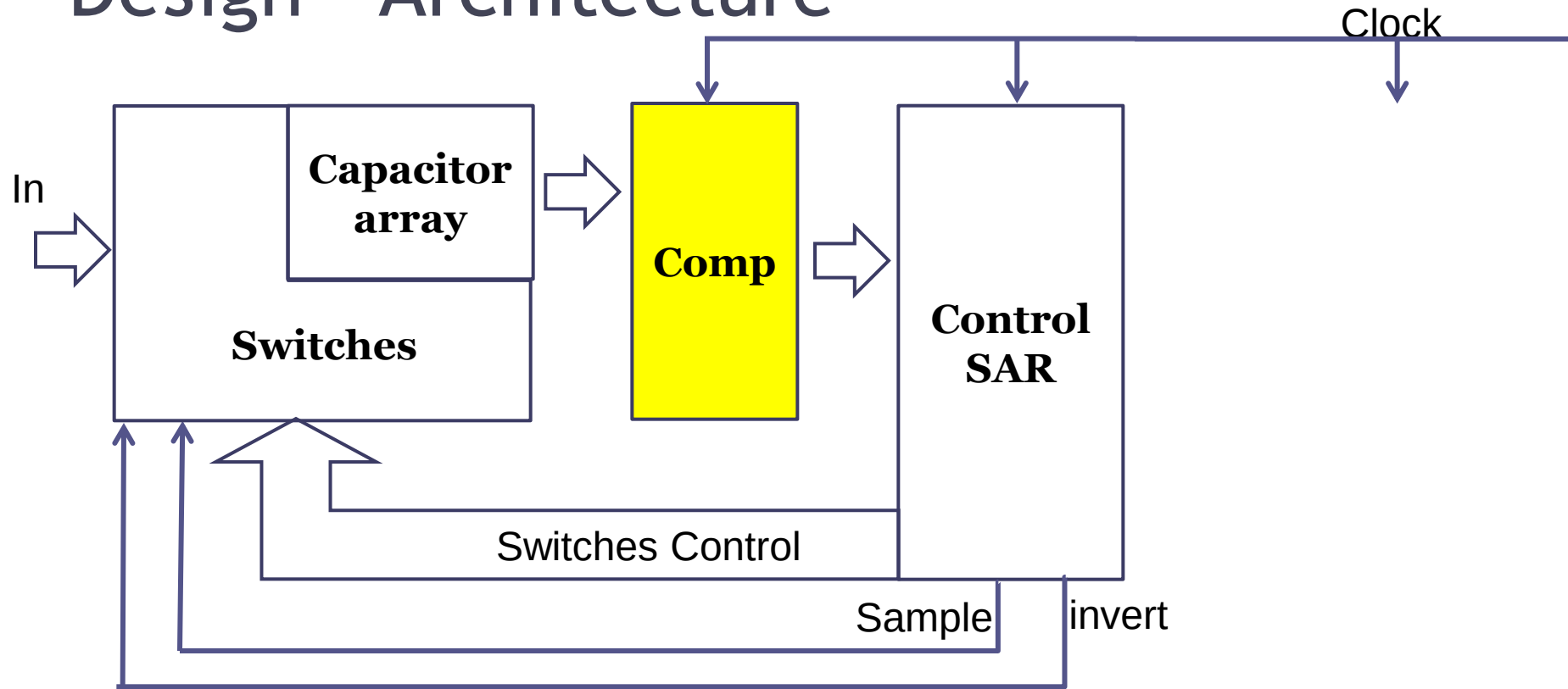
- NMOS to switch V_{gnd} and V_{cm}
- PMOS to switch V_{dd}
- CMOS to switch V_{in}
- Bootstrap to force deep off-state of critical switches



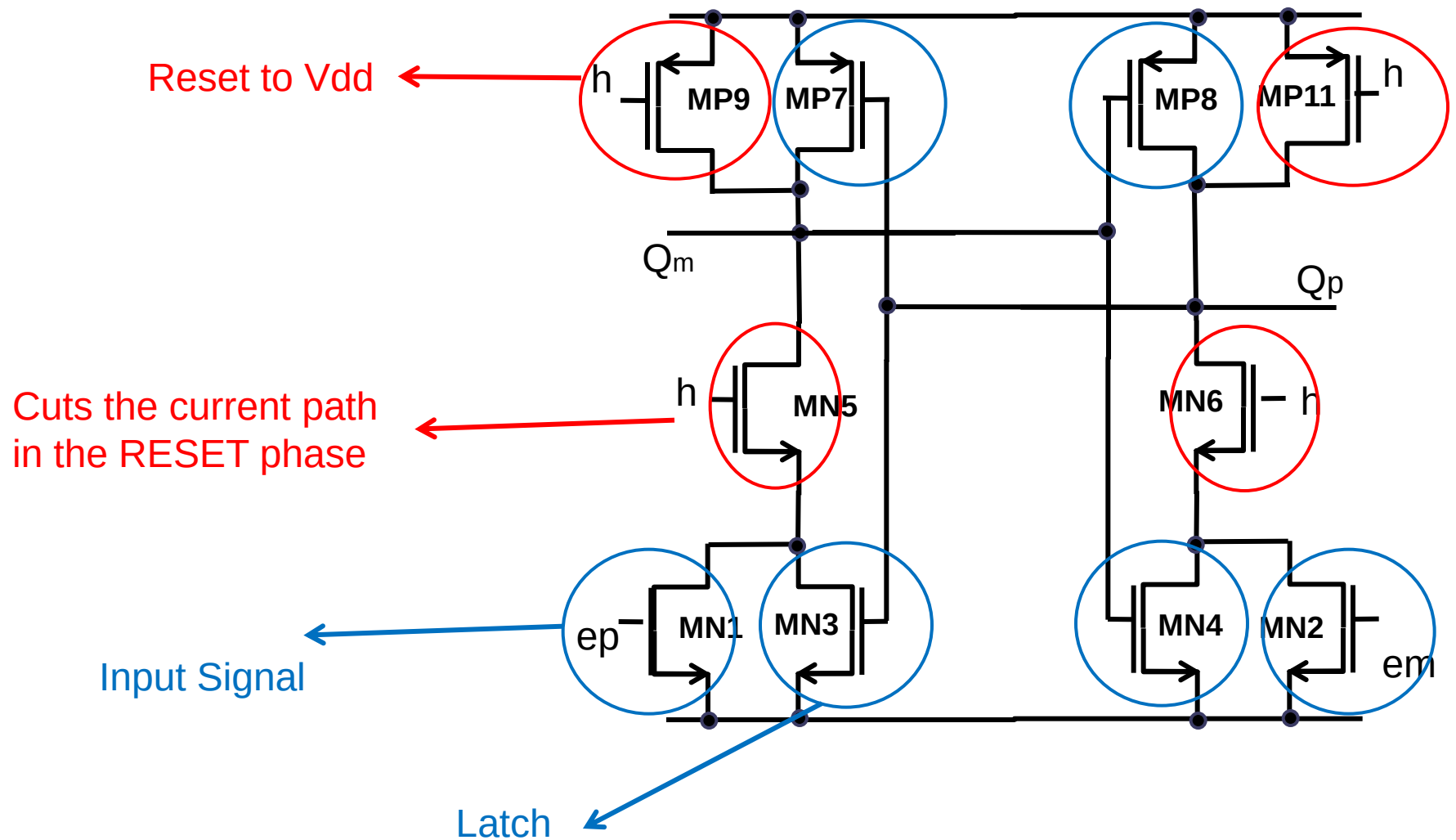
2) Sizing switches (compromise)

- Increasing W/L reduces R_{switch} and so τ , on the account of increasing switch parasitics.
- In the used DAC, this will be of minor importance if operating in low frequency because the switches are all connected to the bottom plates

Design - Architecture



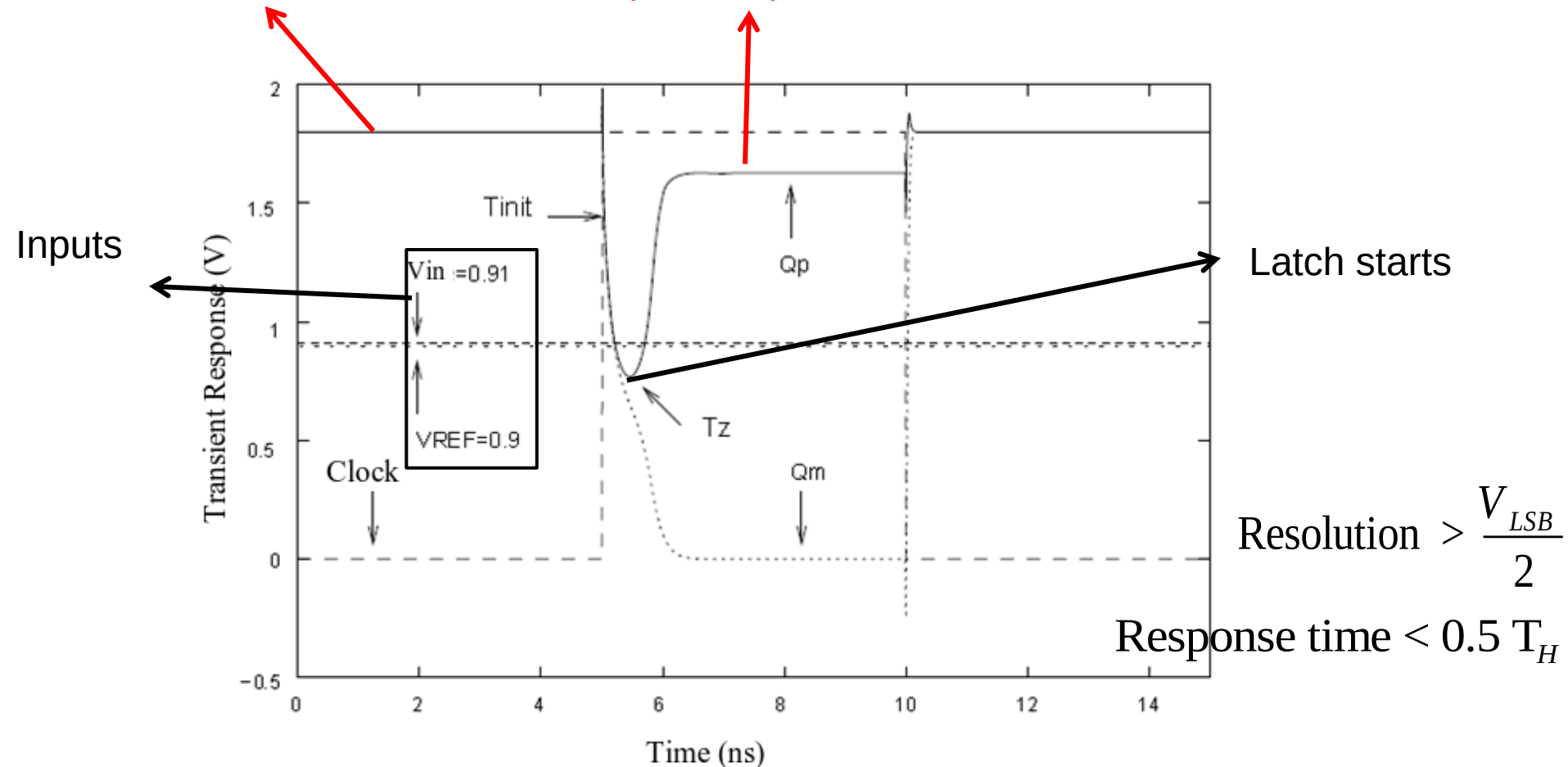
Comparator Circuit



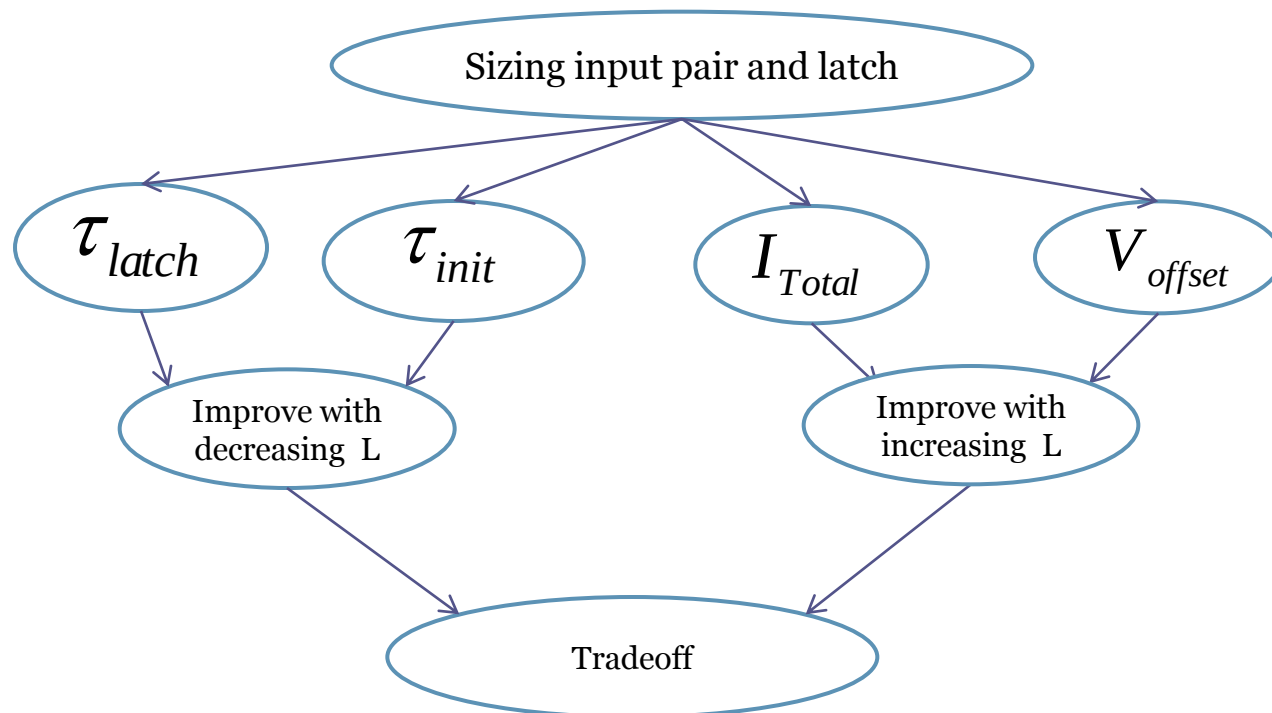
Comparator - Operation phases

Reset phase to Vdd

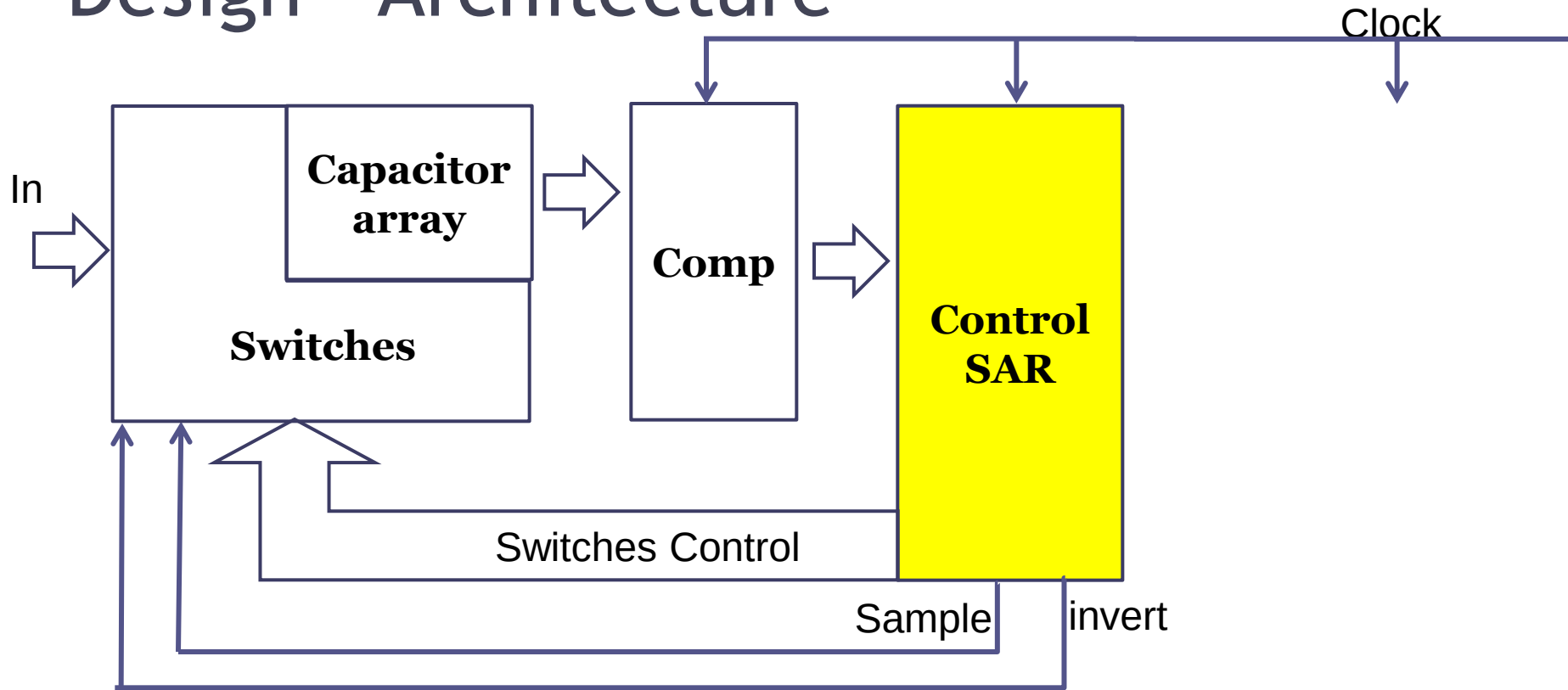
Comparison phase



Comparator - Design tradeoff

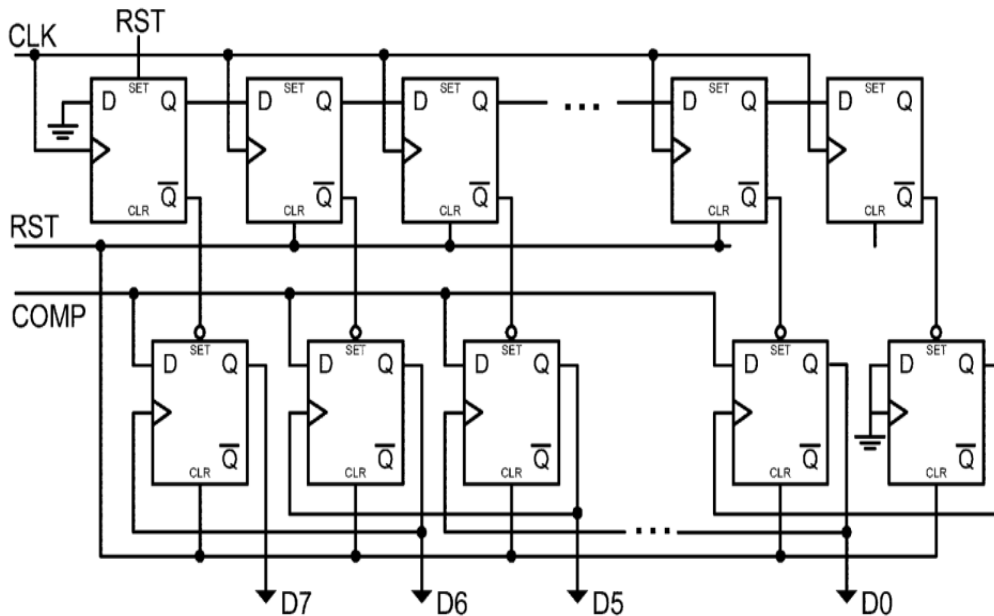


Design - Architecture



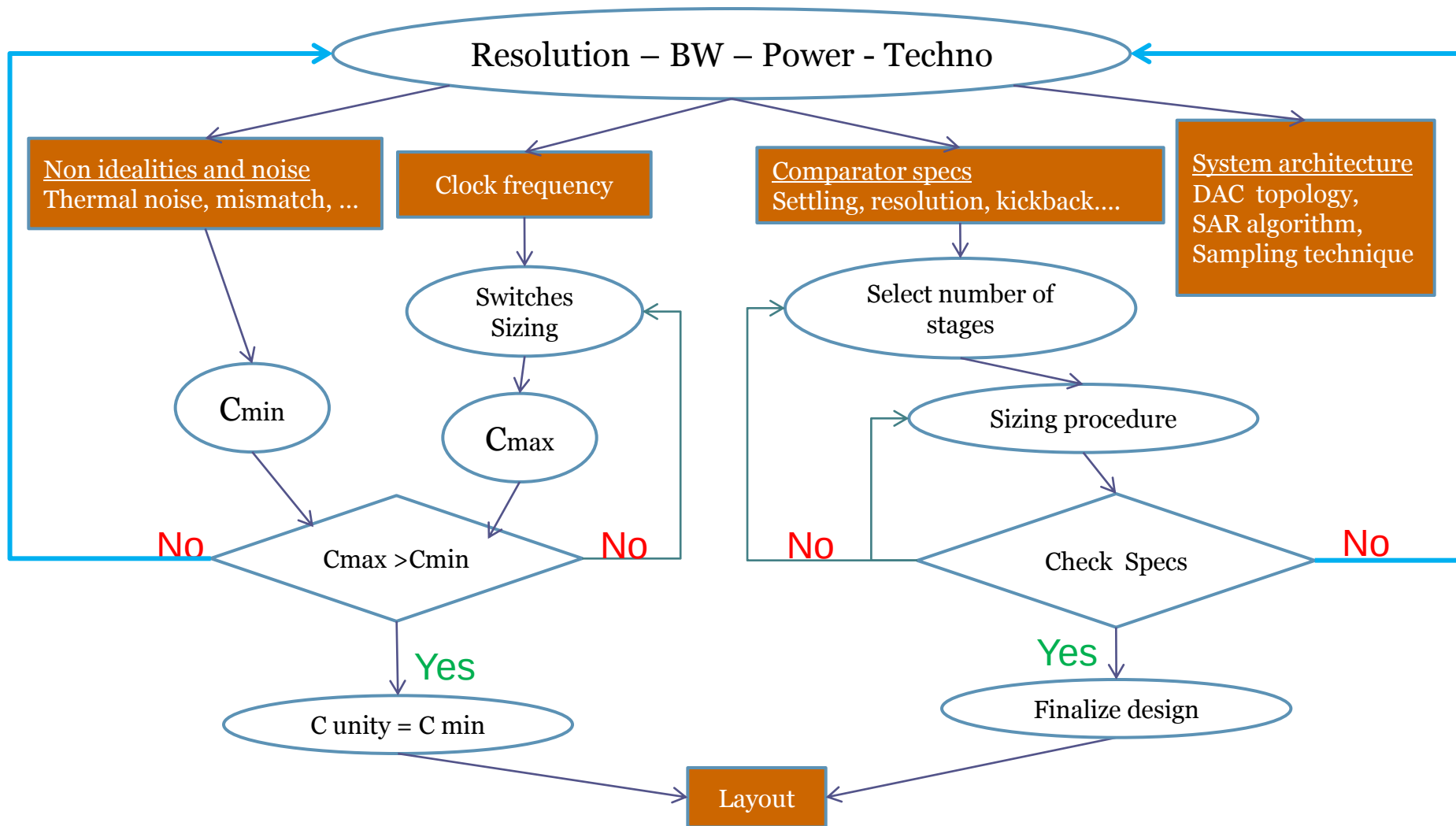
SAR algorithm - Implementation

LFSR: Linear Feedback Shift Register



	DAC outputs								C
0	0	0	0	0	0	0	0	0	X
1	1	0	0	0	0	0	0	0	a8
2	a8	1	0	0	0	0	0	0	a7
3	a8	a7	1	0	0	0	0	0	a6
4	a8	a7	a6	1	0	0	0	0	a5
5	a8	a7	a6	a5	1	0	0	0	a4
6	a8	a7	a6	a5	a4	1	0	0	a3
7	a8	a7	a6	a5	a4	a3	1	0	a2
8	a8	a7	a6	a5	a4	a3	a2	1	a1

Systematic design for SA-ADC



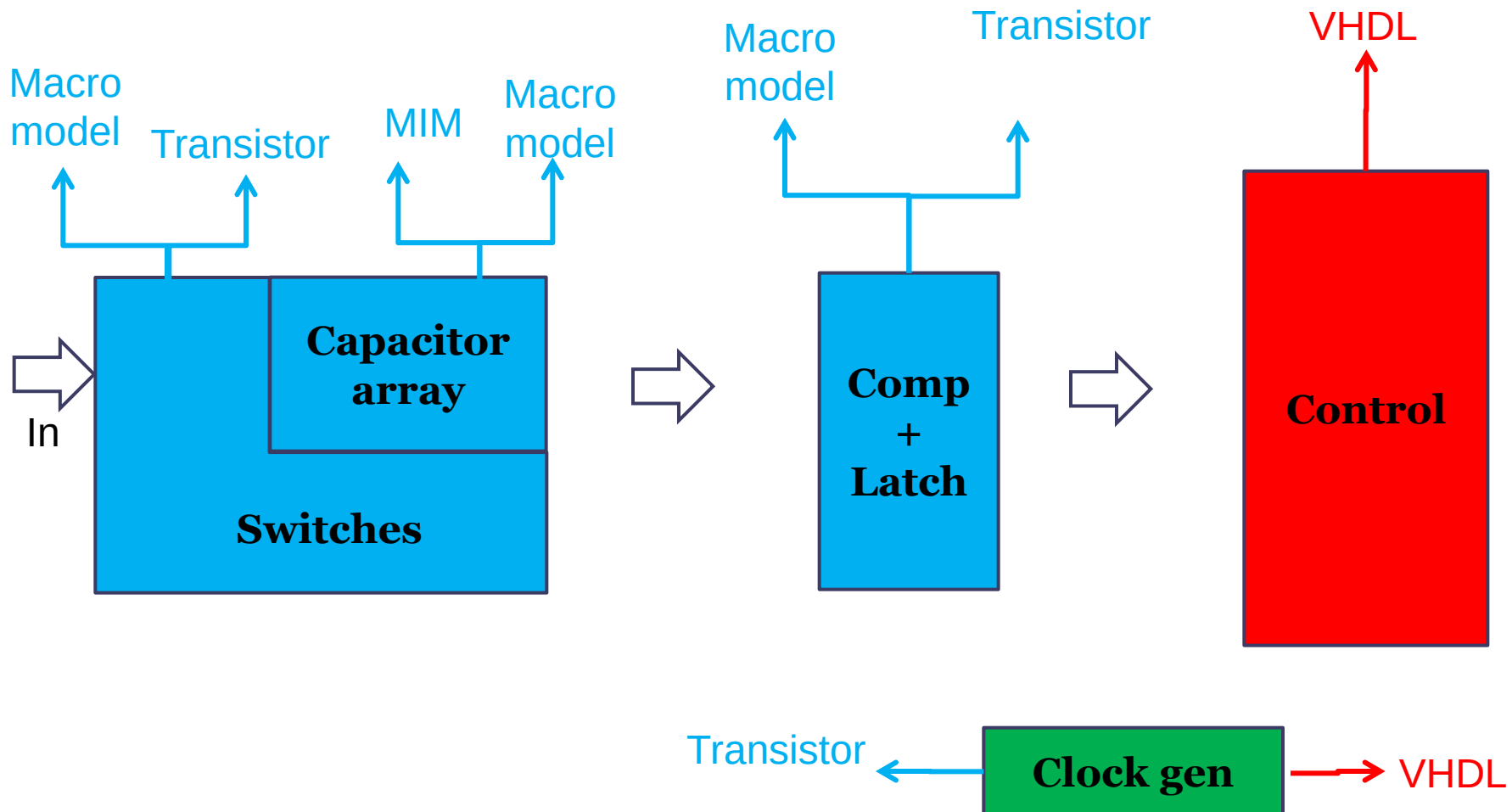
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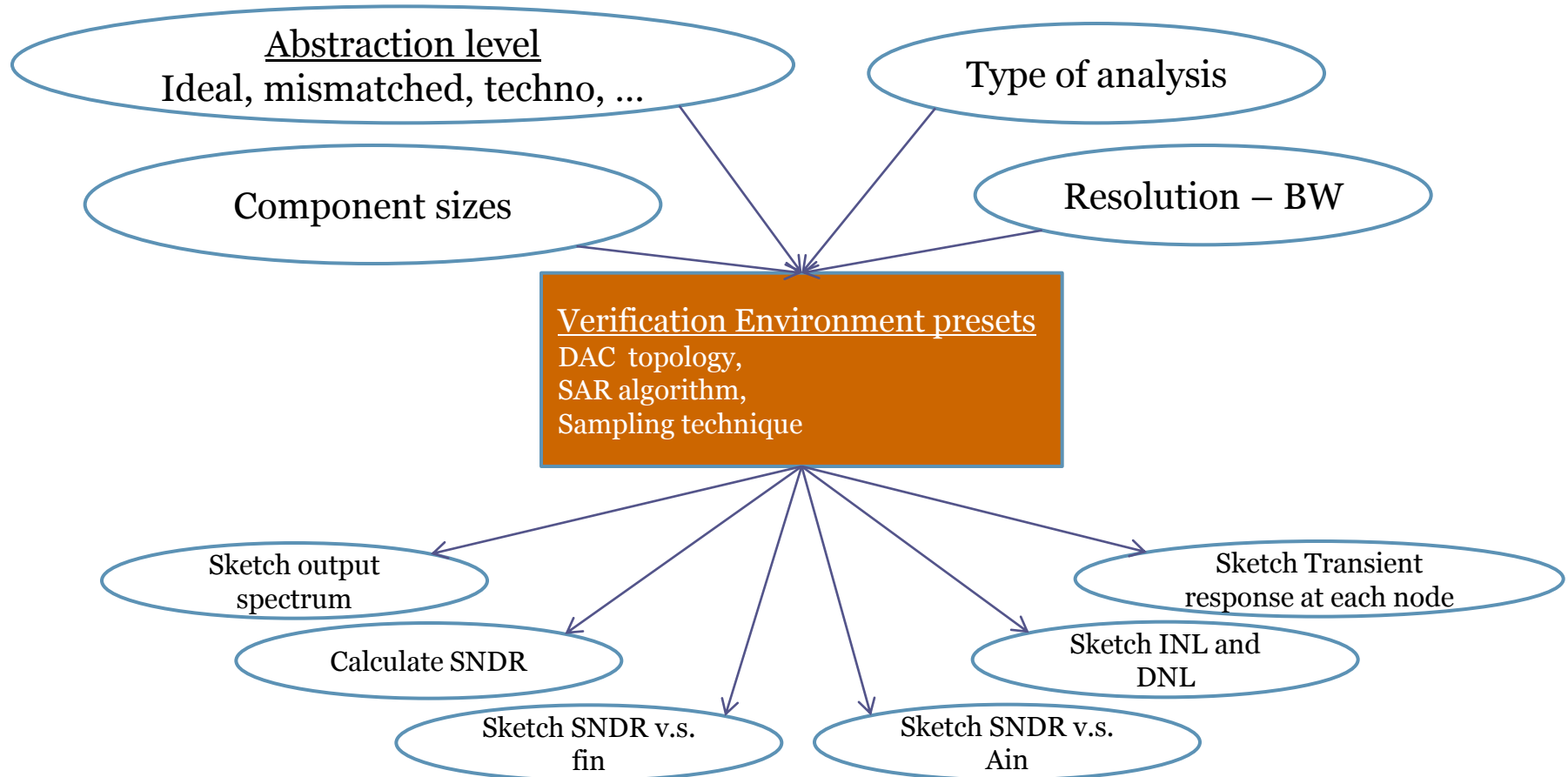
Case Study

- Case Study
 - Differential Architecture
 - Resolution: 8bit
 - BW: 50 KHz
 - F_{clock} : 1MHz
 - Technology: 0.13u ST, MIM Capacitors
- Verification
 - **VHDL AMS** used for verification with simulation
 - Different levels of abstraction (Behavioral , gates, transistor, ...)
 - Mixed blocs simulation (Analog / Digital)

Multiple abstractions



Verification Environment



Transient - Single Ended - Output

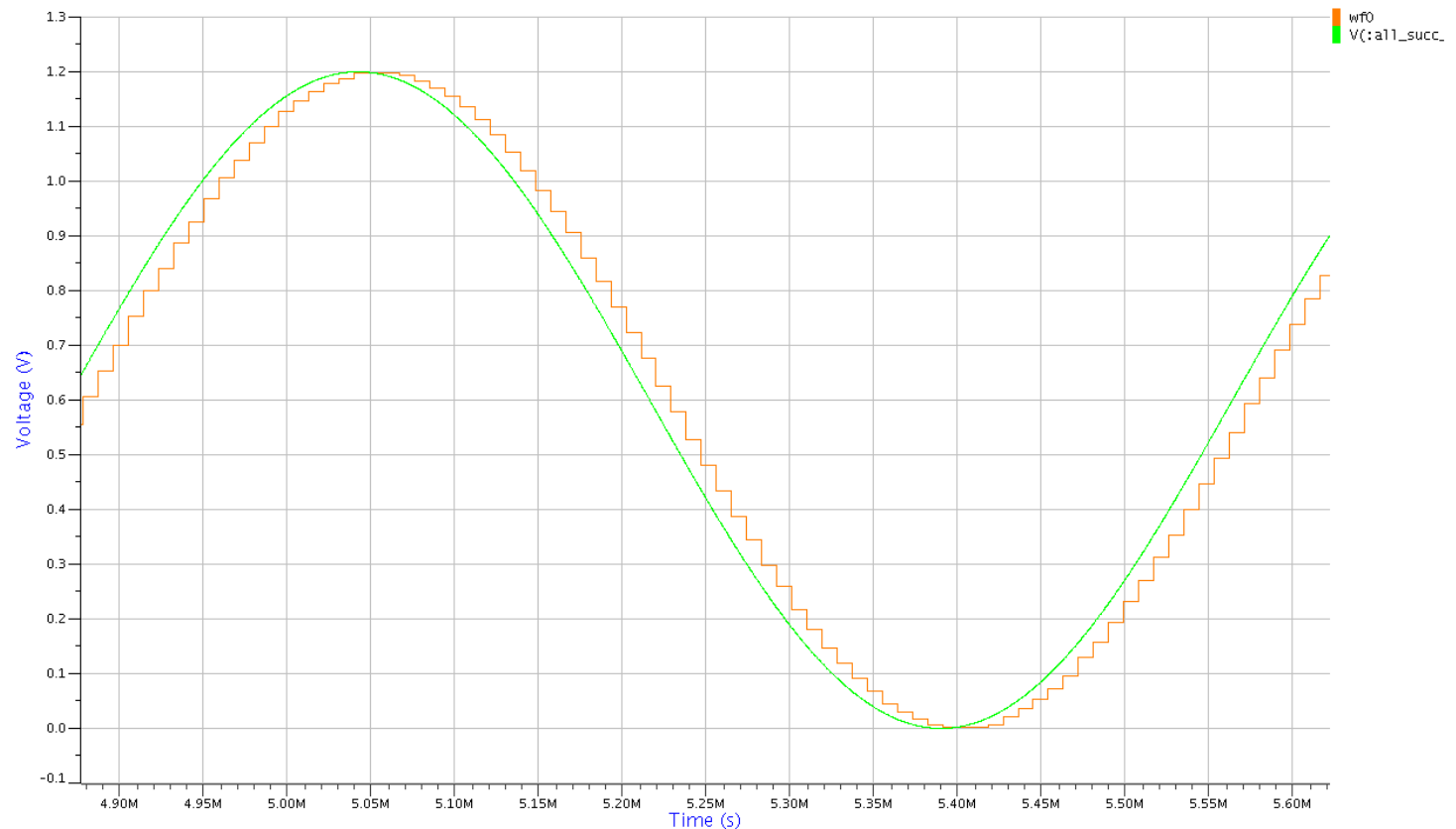
Vdd 1.2V

Fin 1.4KHz

Fclk 1MHz

Vin_{p-p} 1.2V

Transistor level simulations



Transient - Differential - Output

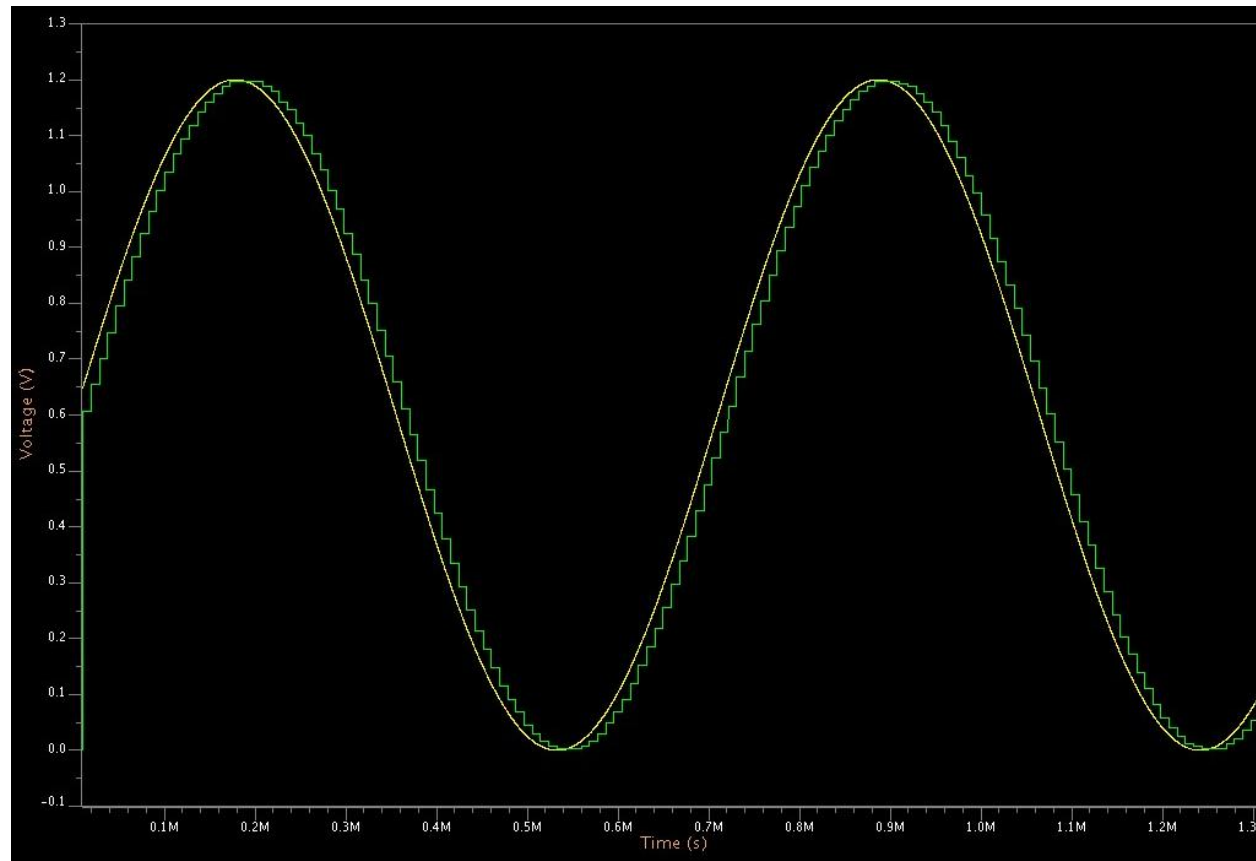
Transistor level simulations

Vdd 1.2V

Fin 1.4KHz

Fclk 1MHz

Vin_{p-p} 1.2V



Transient - Differential - DACs

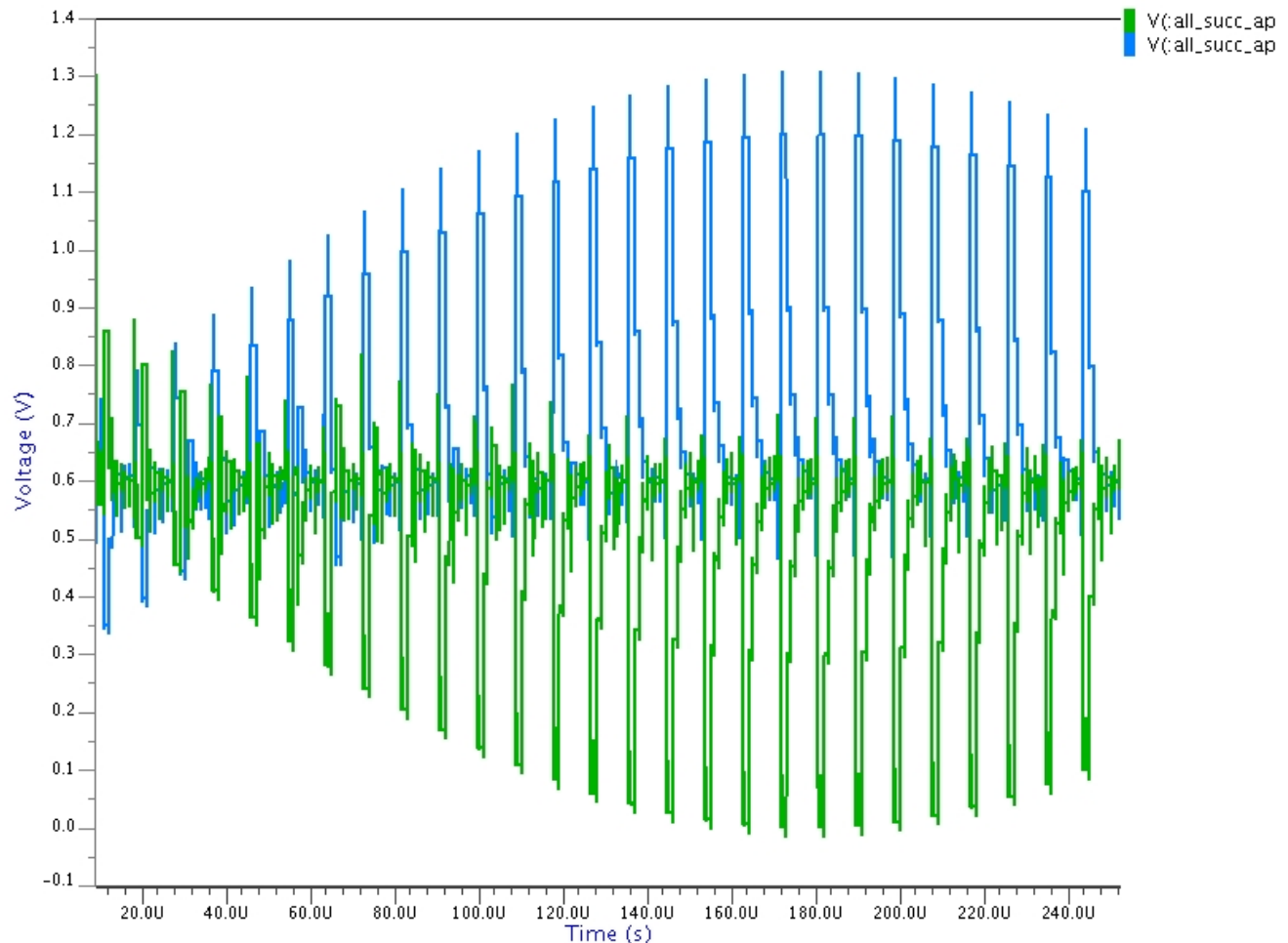
Differential DACs output Transistor level simulations

Vdd 1.2V

Fin 1.4KHz

Fclk 1MHz

Vin_{p-p} 1.2V



Transient - Differential - Comparator

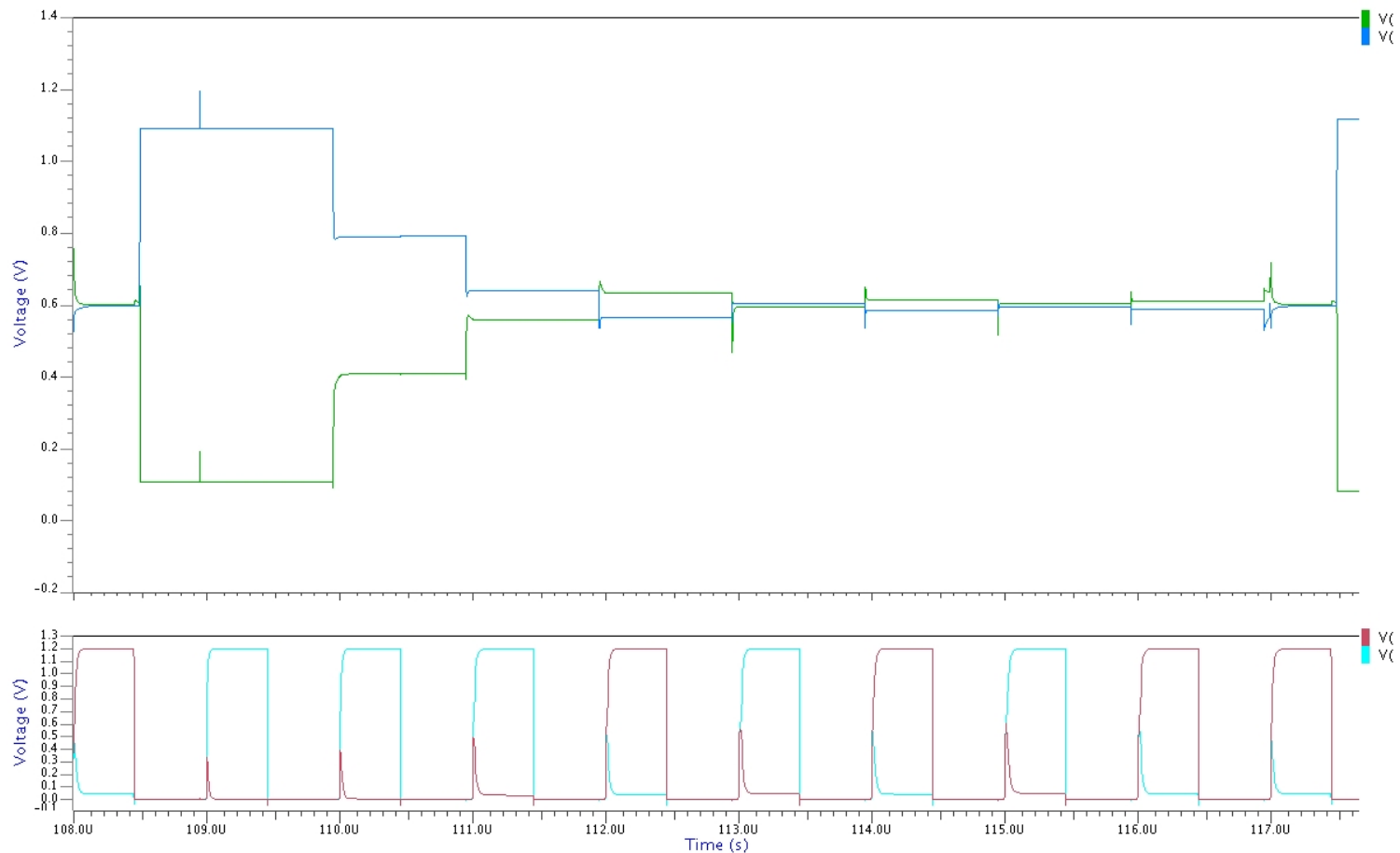
Full conversion: Differential DACs output – Comparator output
Transistor level simulations

Vdd 1.2V

Fin 1.4KHz

Fclk 1MHz

Vin_{p-p} 1.2V



Transient - SAR control

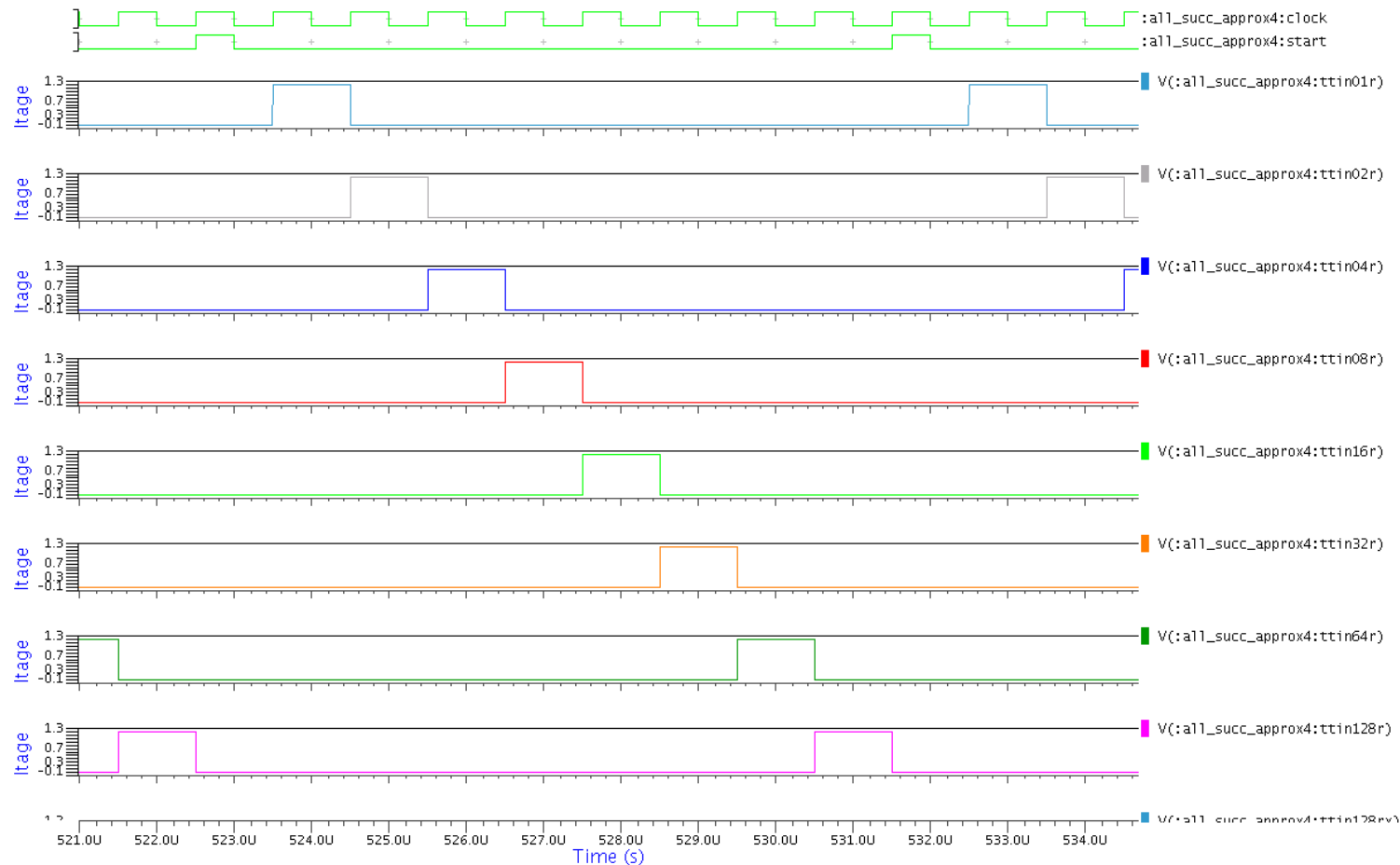
Control block turning ON and OFF DAC switches [case of 0 input]
VHDL Description

Vdd 1.2V

Fin 1.4KHz

Fclk 1MHz

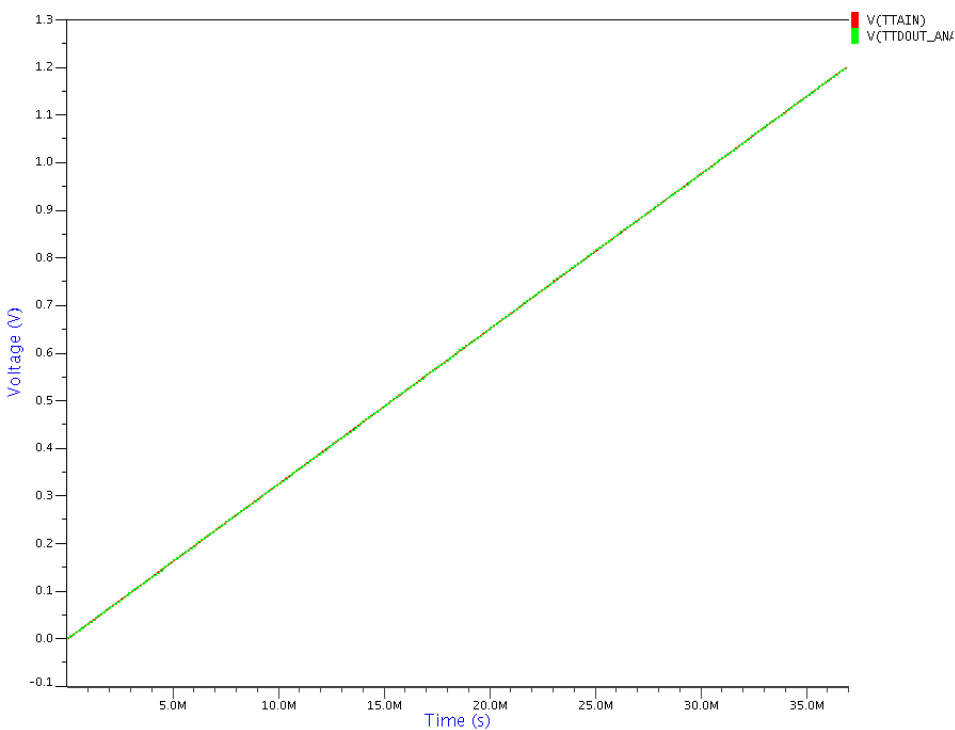
Vin_{p-p} 1.2V



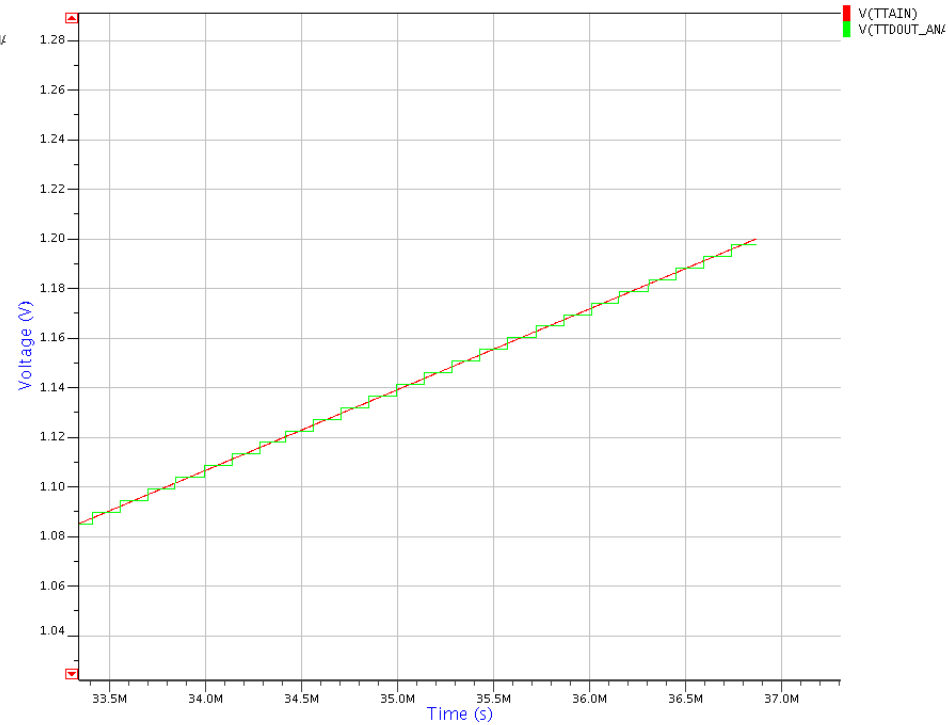
Transient - Full Scale Ramp

Full Scale Slow ramp excitation

Zoom - in



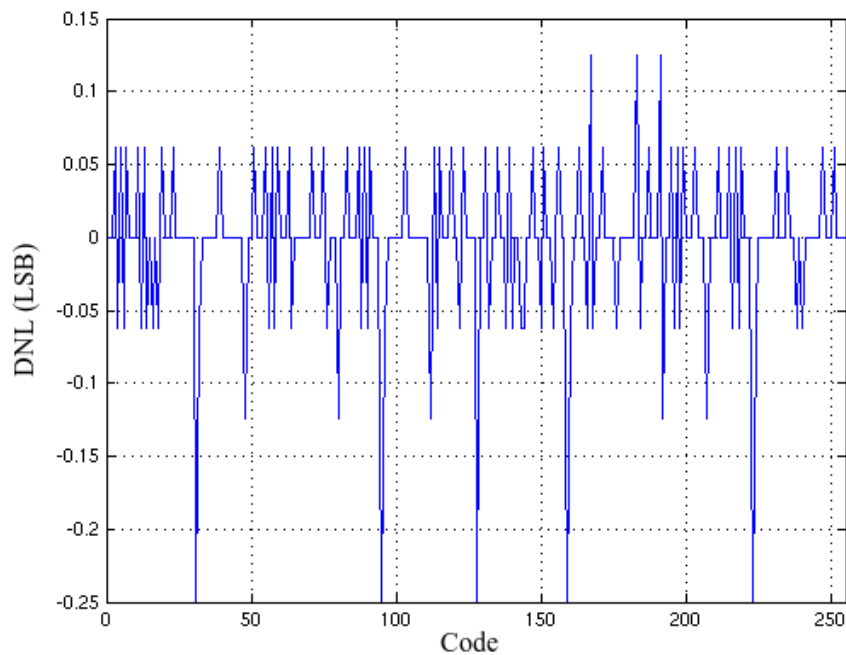
(a)



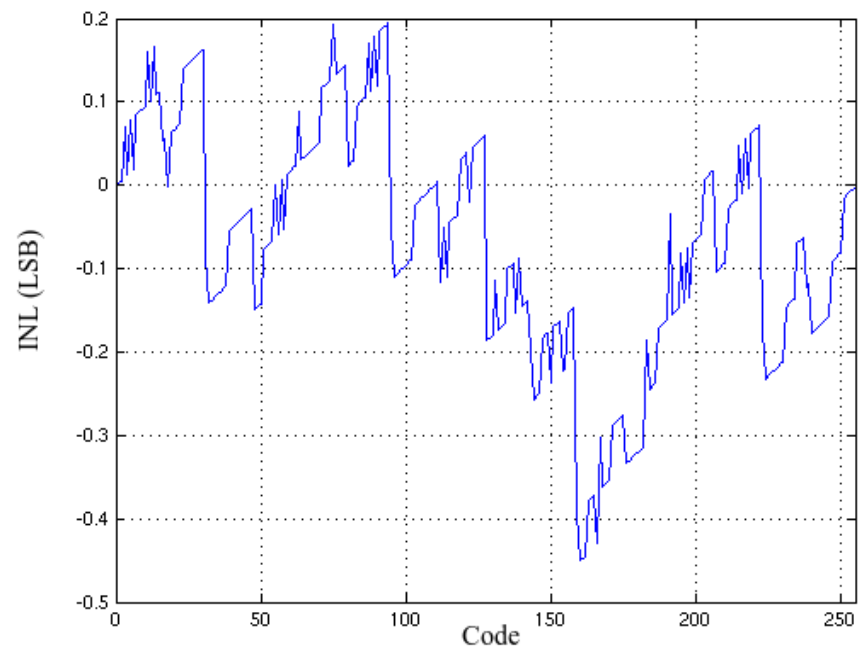
(b)

Static Performance - Transistor Level

Static performance Evaluation in (LSB): DNL and INL [16 sample / bin]



(a)



(b)

Dynamic Performance - Ideal Models

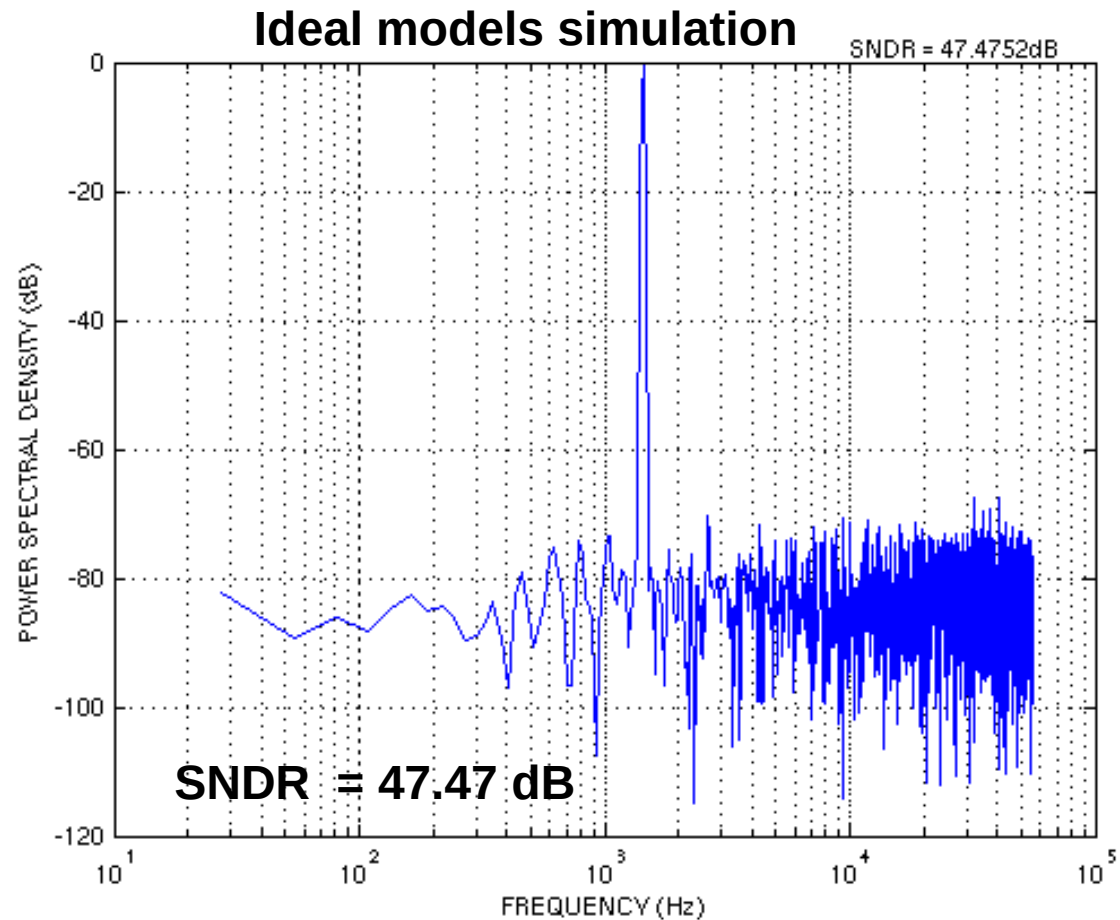
Vdd 1.2V

Fin 1.4KHz

Fclk 1MHz

Vin_{p-p} 1.2V

4096 point FFT



Dynamic Performance - Mixed Models

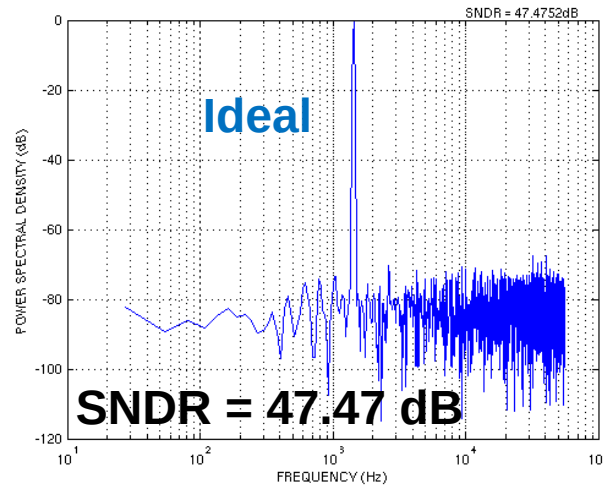
Vdd 1.2V

Fin 1.4KHz

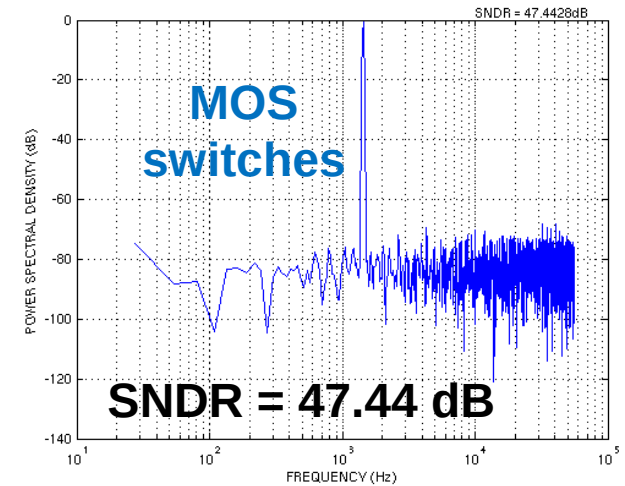
Fclk 1MHz

Vin_{p-p} 1.2V

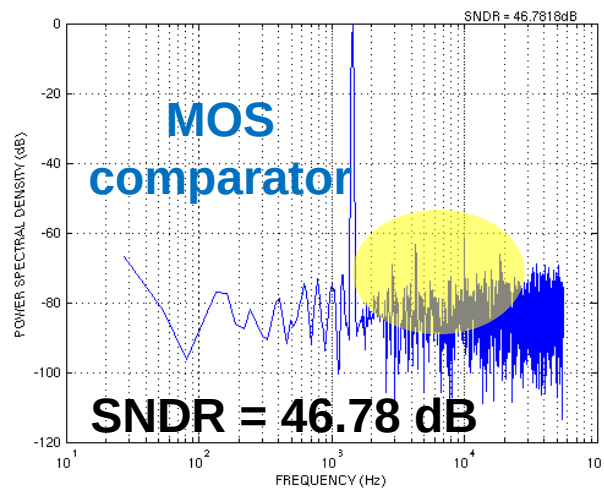
4096 point FFT



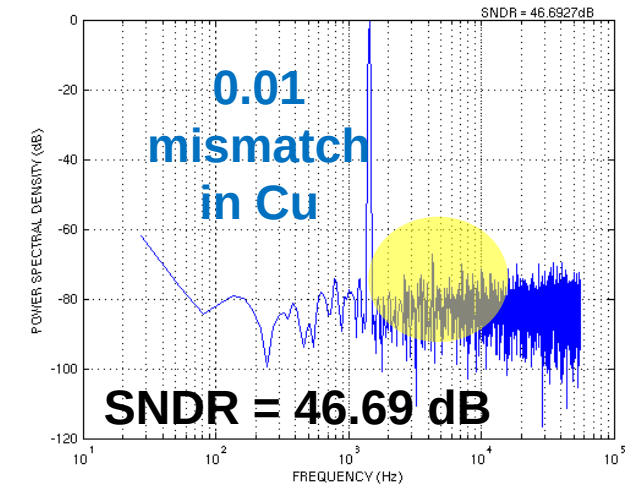
(a)



(b)



(c)



(d)

Dynamic Performance - Transistor Level

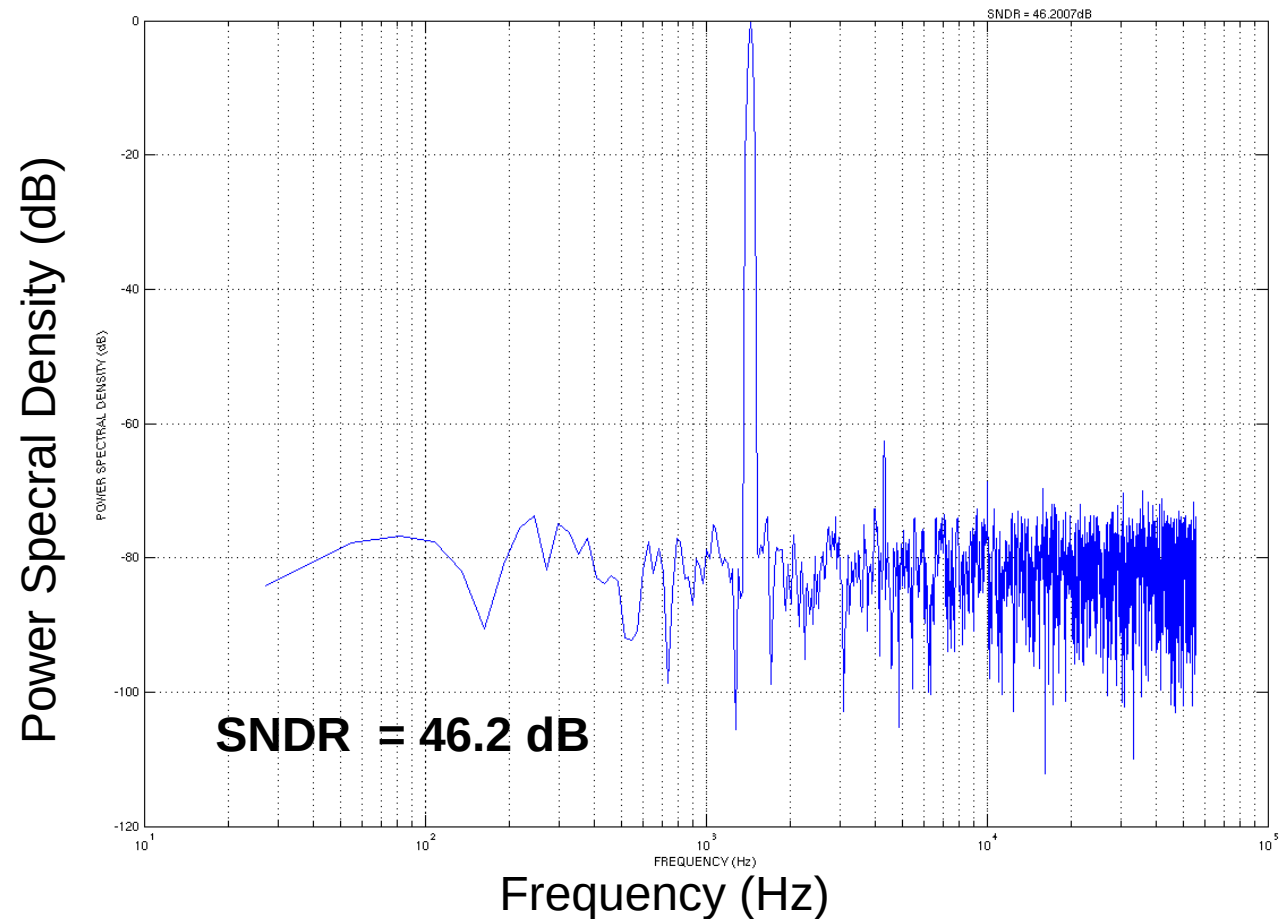
Vdd 1.2V

Fin 1.4KHz

Fclk 1MHz

1024 point FFT

Transistor level simulations



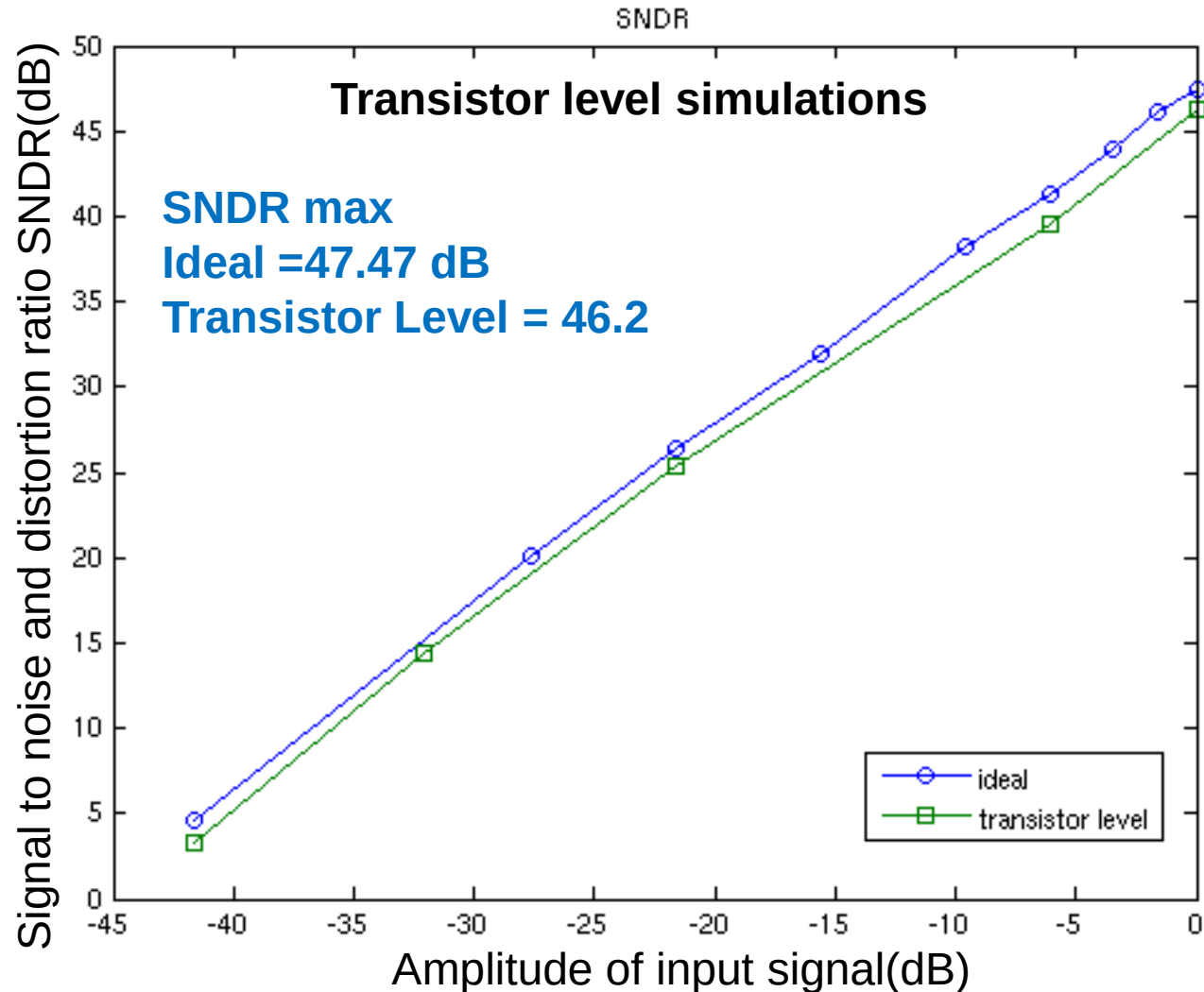
Dynamic Performance

Vdd 1.2V

Fin 1.4KHz

Fclk 1MHz

4096 point FFT



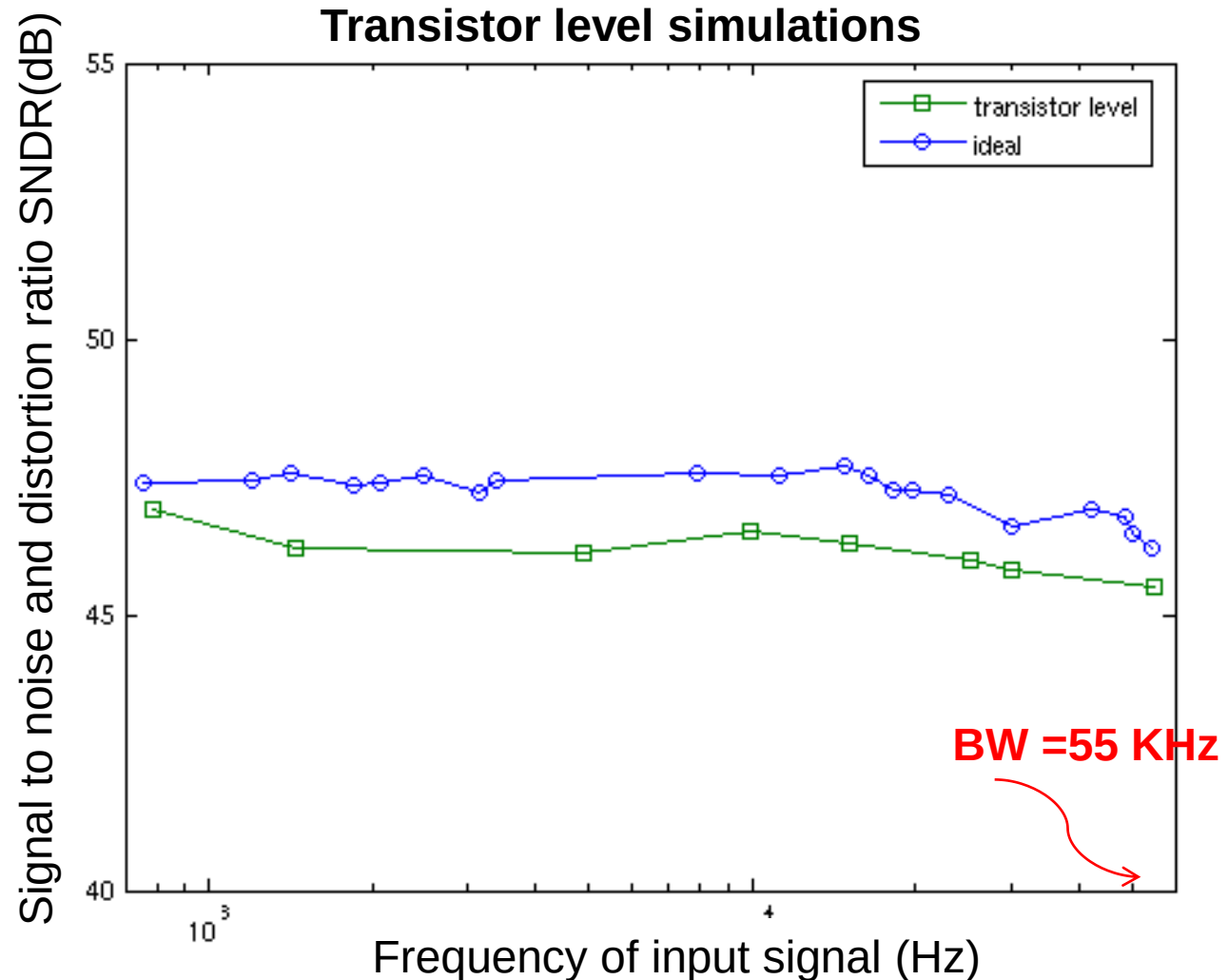
Dynamic Performance

Vdd 1.2V

Fclk 1MHz

Vin_{p-p} 1.2V

4096 point FFT



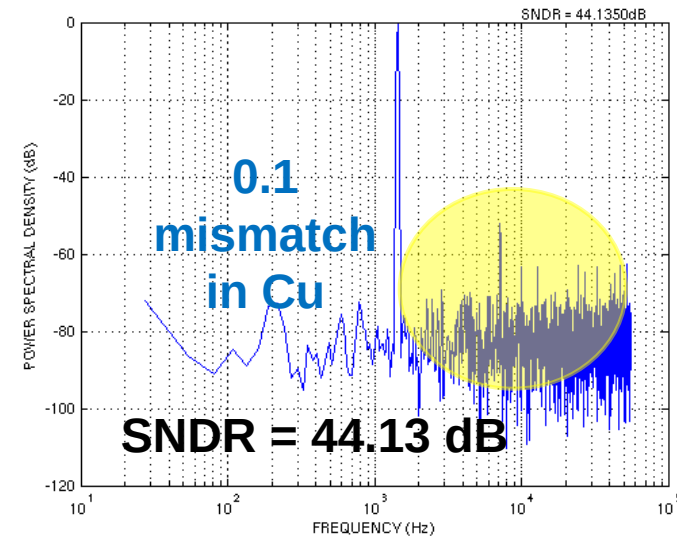
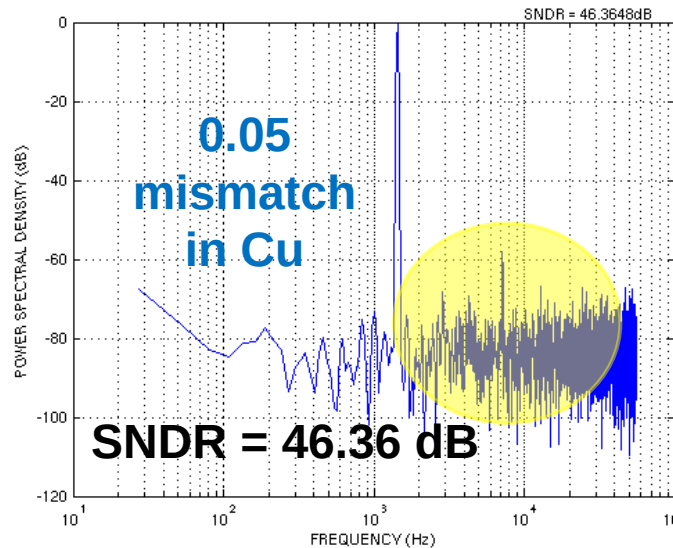
Mismatch analysis

Vdd 1.2V

Fin 1.4KHz

Fclk 1MHz

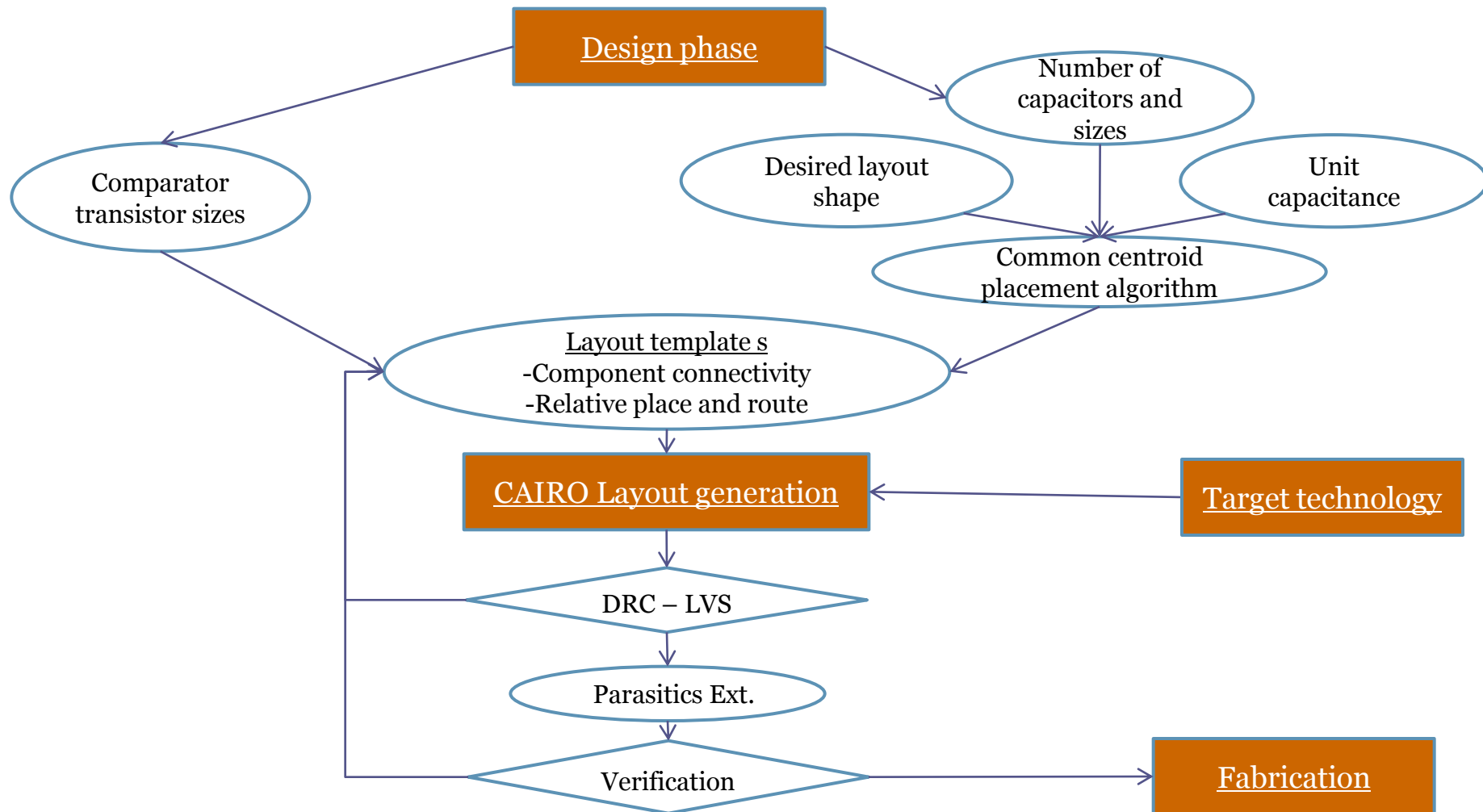
Vin_{p-p} 1.2V



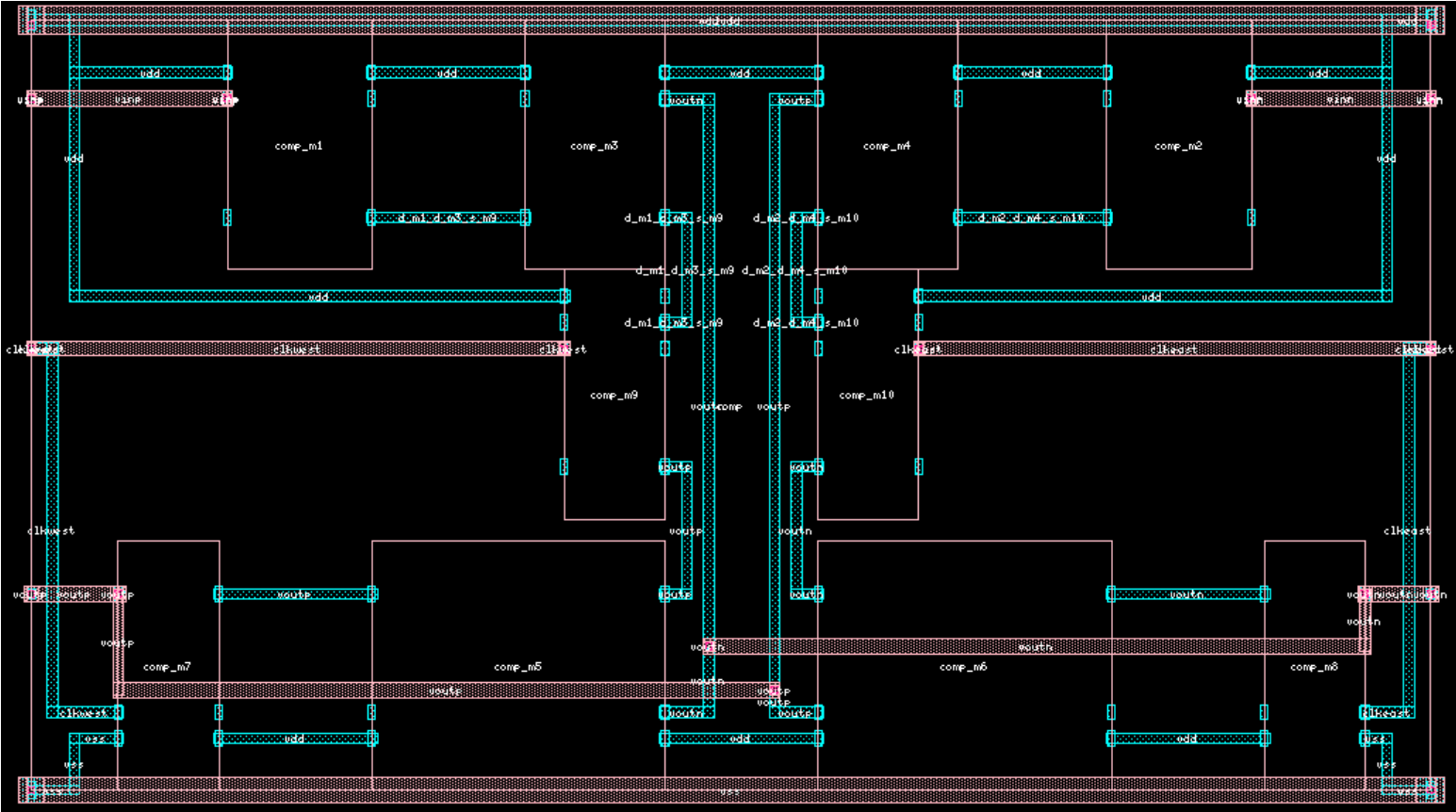
4096 point FFT

C_u	Variance σ					Ideal
	0.01	0.02	0.05	0.1	0.2	
10fF	47.36	46.40	43.22	39.10	35.6	47.44
20fF	47.2	47.09	44.99	42.08	35.77	47.44
30fF	43.89	44.6	40.19	39.02	32.7	47.44

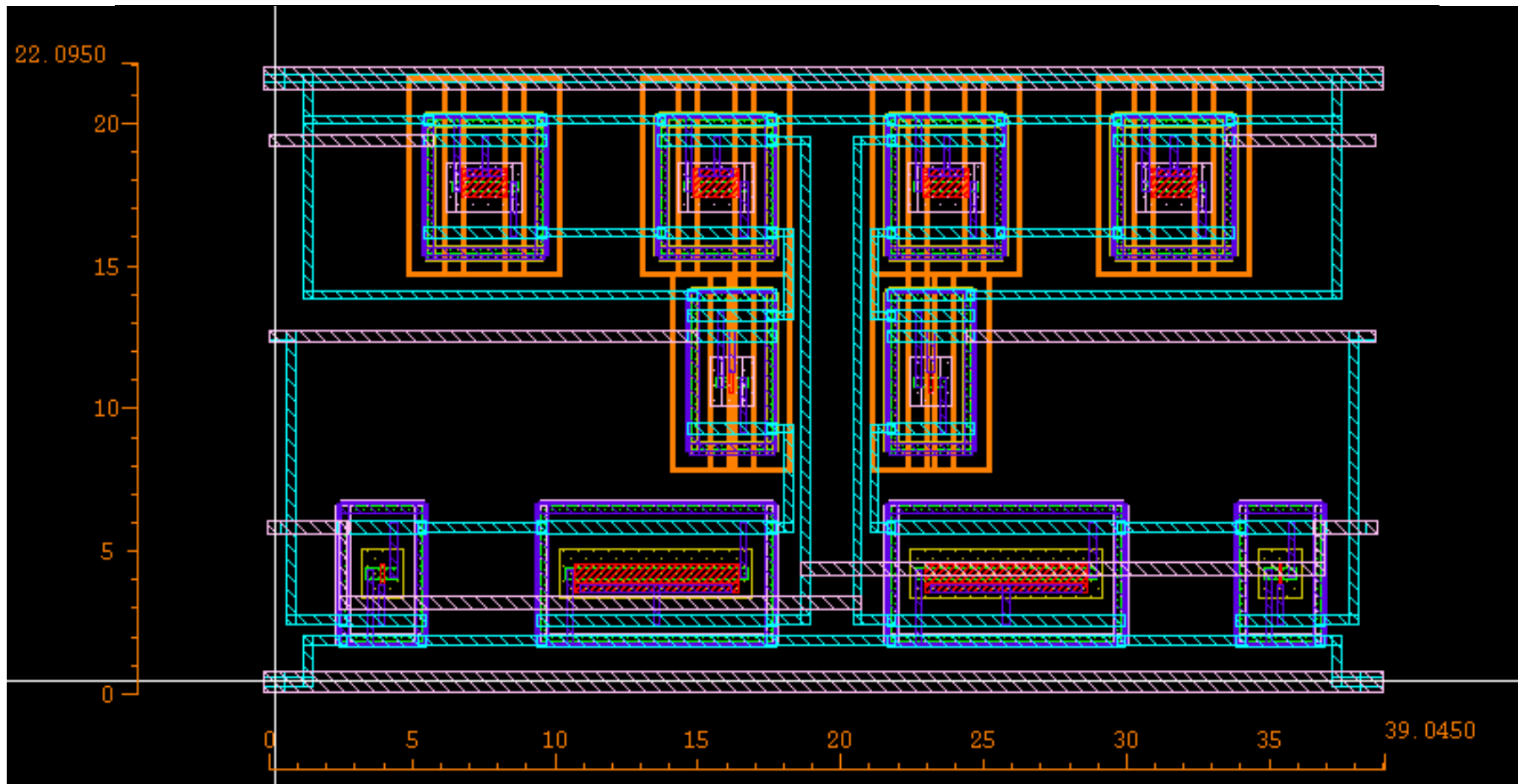
Layout generation for SA-ADC



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Layout - Comparator - Generated



Area 22 x 39 μm^2

Dummies Removed for Layout verification

Layout - Differential DACs - Floorplan

Common centroid placement for 16 capacitor

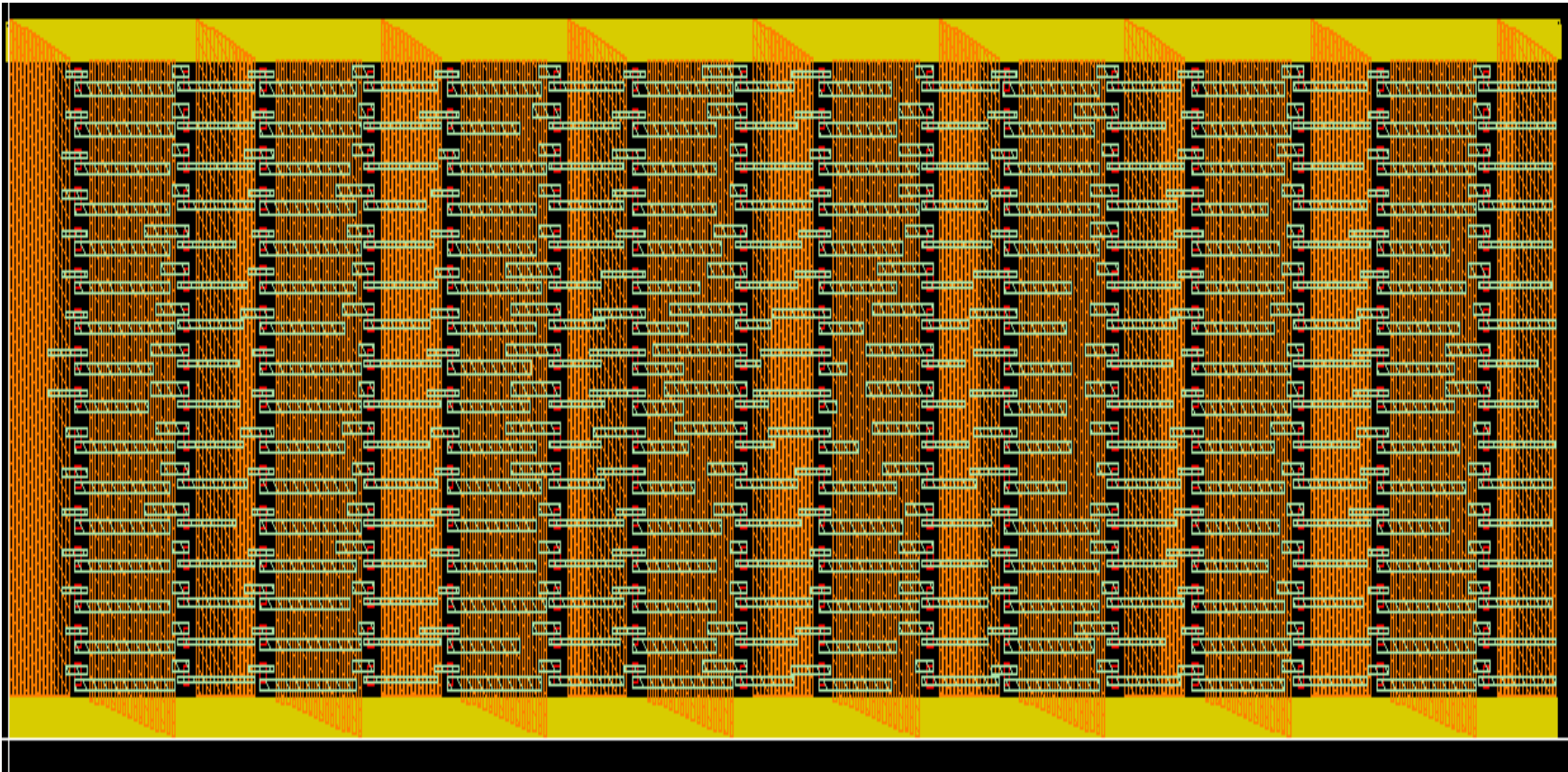
```

how many capacitors? 16
routing? 1
enter values
1 1 1 1 2 2 4 4 8 8 16 16 32 32 64 64
use rectangles?(1=yes)1
enter unit capacitor side length:
1
enter length:
16
sum:256
16 16 15 15 15 15 16 11 11 16 15 15 15 15 16 16
16 16 16 16 11 14 13 12 12 13 14 11 16 16 16 16
15 16 14 15 16 15 13 16 16 13 15 16 15 14 16 15
15 16 15 12 14 15 13 16 16 13 15 14 12 15 16 15
15 11 16 14 14 15 14 13 13 14 15 14 14 16 11 15
15 14 15 15 15 9 10 12 12 9 10 15 15 15 14 15
16 13 13 13 14 10 8 5 6 8 9 14 13 13 13 16
11 12 16 16 13 9 7 2 3 7 10 13 16 16 12 11
11 12 16 16 13 10 7 4 1 7 9 13 16 16 12 11
16 13 13 13 14 9 8 6 5 8 10 14 13 13 13 16
15 14 15 15 15 10 9 12 12 10 9 15 15 15 14 15
15 11 16 14 14 15 14 13 13 14 15 14 14 16 11 15
15 16 15 12 14 15 13 16 16 13 15 14 12 15 16 15
15 16 14 15 16 15 13 16 16 13 15 16 15 14 16 15
16 16 16 16 11 14 13 12 12 13 14 11 16 16 16 16
16 16 15 15 15 15 16 11 11 16 15 15 15 15 16 16

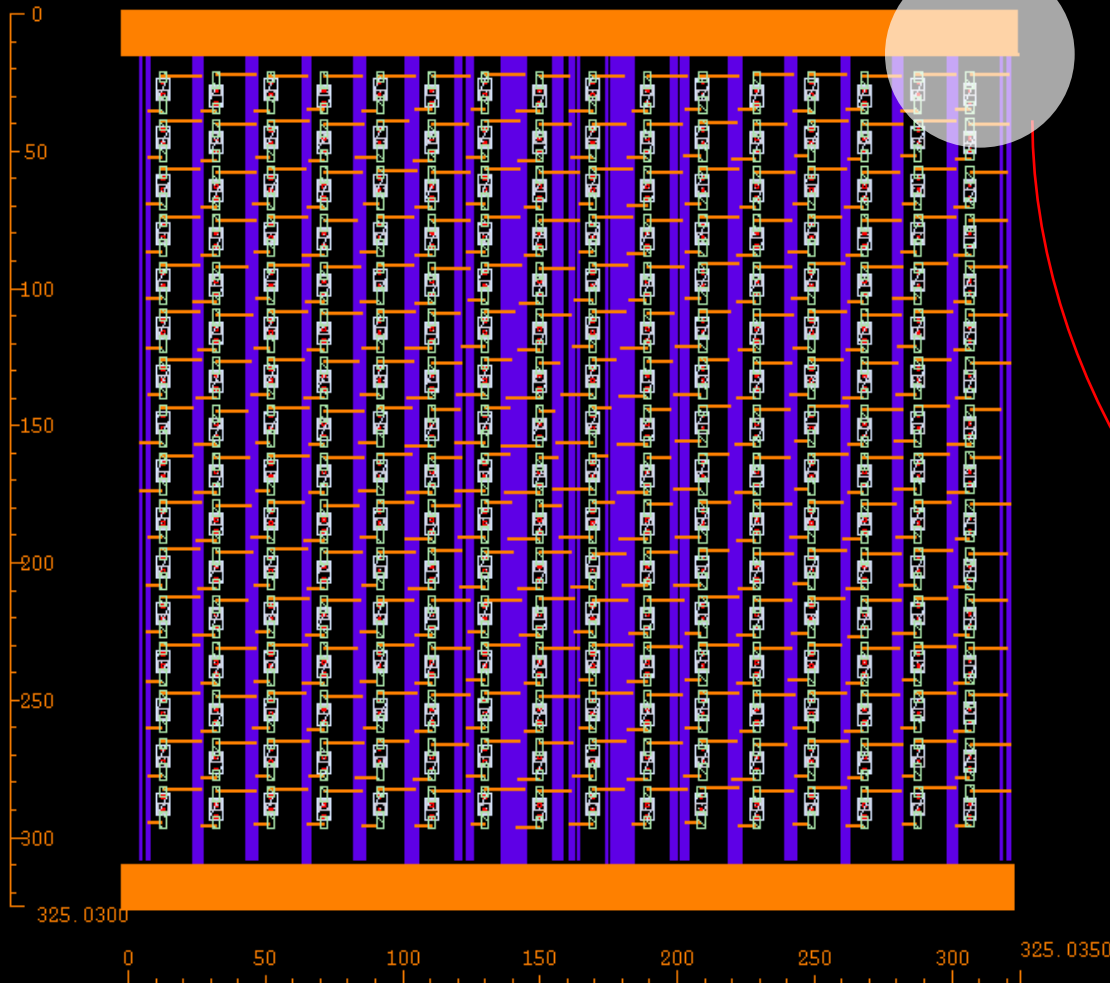
```

Layout - Differential DACs - Generated

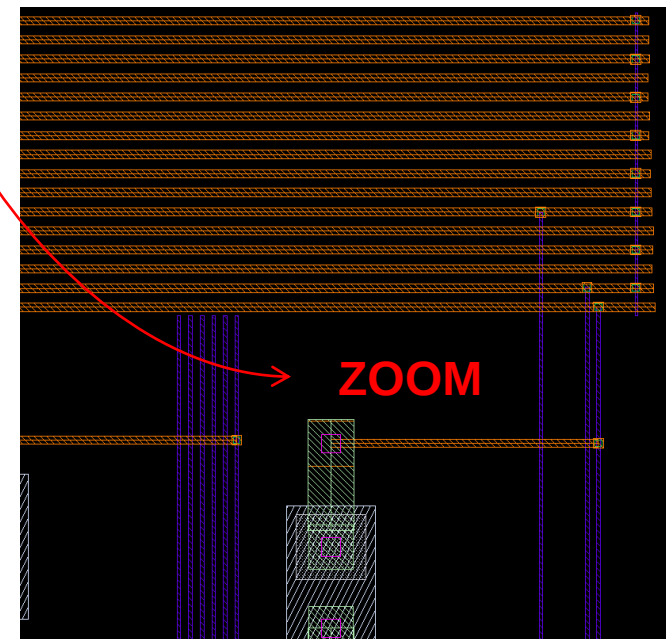
Layout 1 - 256 Cu – Placed and Routed – Area
1.26 x 0.26 mm² and Huge routing parasitics



Layout - Differential DACs - Manual



Layout 2 - 256 C_u – Placed
and Routed – 2/3 less
routing parasitics
Area 0.1056 mm²



Performance

	[Hongo7]	This work*	[scotto3]
Technology	0.18 μm	0.13 μm	0.25 μm
Supply	0.83 V	1.2 V	1.0 V
Input range	Rail to Rail	Rail to Rail	Rail to Rail
Sampling rate	111 KHz	111 KHz	100 KHz
Unit Cap.	24 fF	30fF	12f
Power (Analog)	1.16 μW	0.72 μW	2.2 μW
Area	0.062 mm ²	0.122 mm ²	0.053 mm ²
SNDR@BW	47.40 dB	46.2dB	43.8 dB
Architecture	Single Ended	Differential	Single Ended
FOM	65 fJ/bit	64fJ/bit	2163 fJ/bit

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Summary and Conclusion

- Systematic design methodology for SA-ADC from system to layout.
- General simulation environment
 - Different abstraction levels.
 - Different verification tests.
- Emphasis on analog design automation and reuse
- Optimizing Layout for best component matching
- Verification with case study for WSN specs

Perspectives

- Targeting high frequency specs (>500 Msample/S)
 - Redundant system error correction code [Kuttner02]
 - Digital calibration [Promitzer01]
 - Asynchronous operation [Chen06]
 - Time interleaving [Chen06]
- Full Automation
 - Sizing procedure with layout parasitics awareness
 - Layout generation for the full ADC

Thank You